



Flipsky ESC Tool V1.5
User Manual

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1 Overview

Flipsky ESC Tool is developed based on the FT ESC series, only adaptable for FT ESC series of Flipsky. The main function of this APP includes FT series ESCs and parameters settings, control mode setup, control test and data display, firmware updating, ESC restoration of factory settings, ESC reset and other functions.

Thanks for purchasing FT series ESC! In order to ensure the normal use of the product and avoid possible equipment damage, we recommend that you read this manual carefully before using and strictly follow the specified operating procedures. Our company will not be held responsible for any liability arising from any unauthorized modifications. Meanwhile, our company reserves the right to change the product design, appearance, performance and use requirements without notice.

2 Operation Requirements

2.1 Hardware Requirement

- FT Series ESC

Current Products including: FT85BS, FT85BS2.0, FT85BD, FT85RS, FT60BD, FT85KS, FT85GS。

- Type-C USB cable

2.2 Software Requirement

- Windows 7 or above operation system is required;
- Windows 7 and Windows 8 need to install CDC port driver, or Flipsky ESC Tool cannot build communication with FT series ESC.
- Windows 10 or above have built-in CDC port driver.

2.2.1 System Driver Download and Related Files Description

Join QQ group to download related files: 1039697455。

Related folder including: (as Fig.2-1)

Fig.2-1. Folder content

Name	Date Changed
 Win7 or Win8 CDC Driver	2025/6/28 10:40
 Flipsky ESC Tool V1.5 20250511	2025/5/12 8:45
 Flipsky ESC Tool V1.5 User Manual_...	2025/6/28 10:44

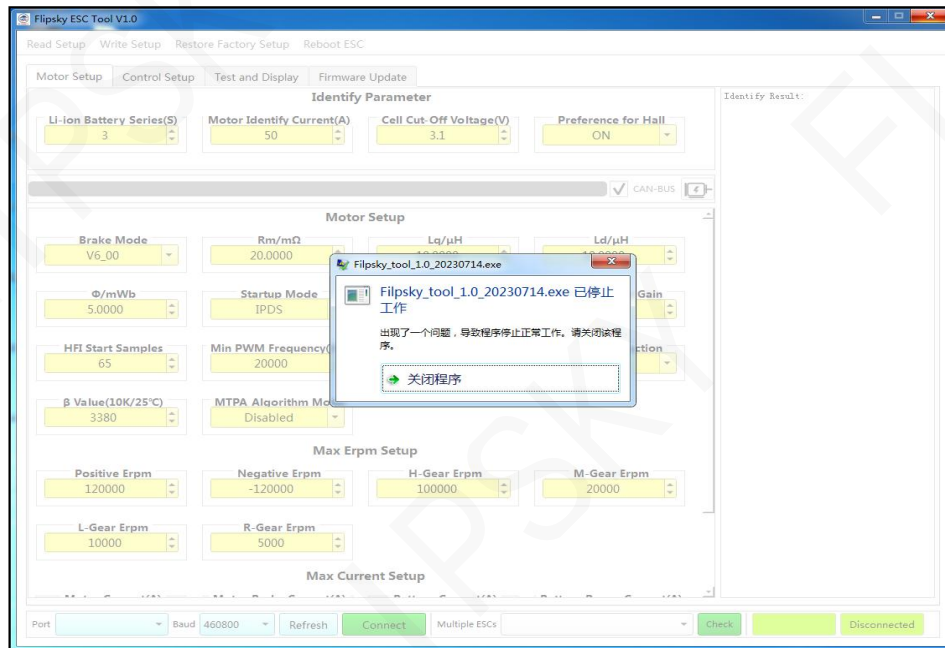
- Win7_or_ Win8 CDC Driver: Window 7 / Window 8 CDC port driver installation folder
- Flipsky ESC Tool V1.5 20250511.exe;
- Flipsky ESC Tool User Manual V1.5.

2.2.2 Windows 7 / Windows 8 Driver Installation

Take Windows 7 as an example:

The ESC might not connect with the ESC Tool for the first time if you use Windows 7. (as Fig.2-2) Please update / reinstall the port driver for your computer system to connect.

Fig.2-2. Fail to connect



Drive Update:

- Connect the ESC with computer via USB cable, right click the "Computer" and select "Device Manager" as Fig.2-3:

Fig. 2-3. Open Device Manager

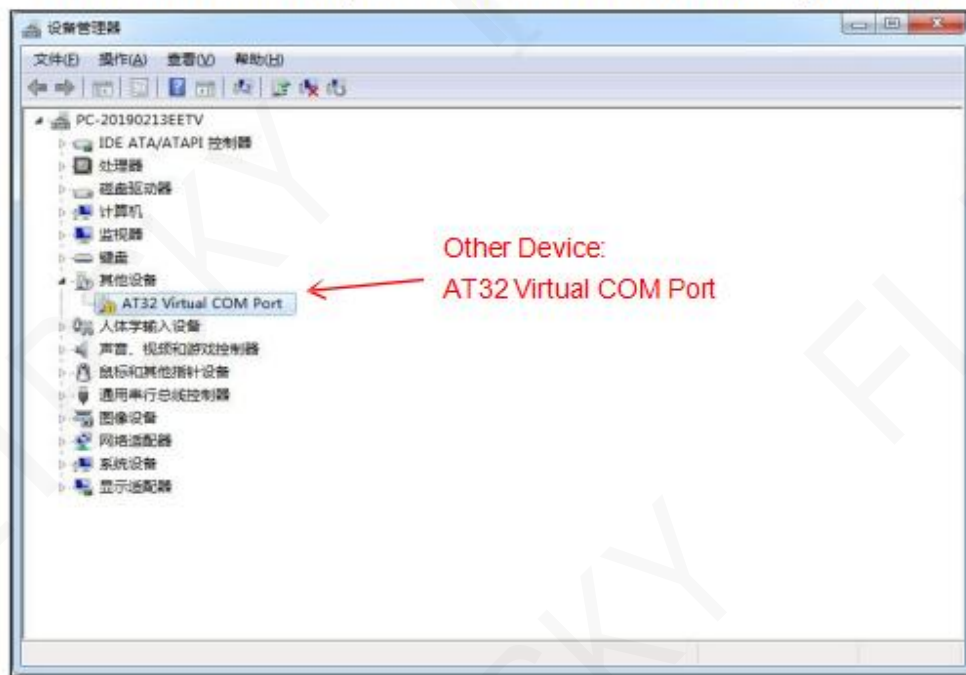
Right-click the "Computer"



Select "Device Manager"

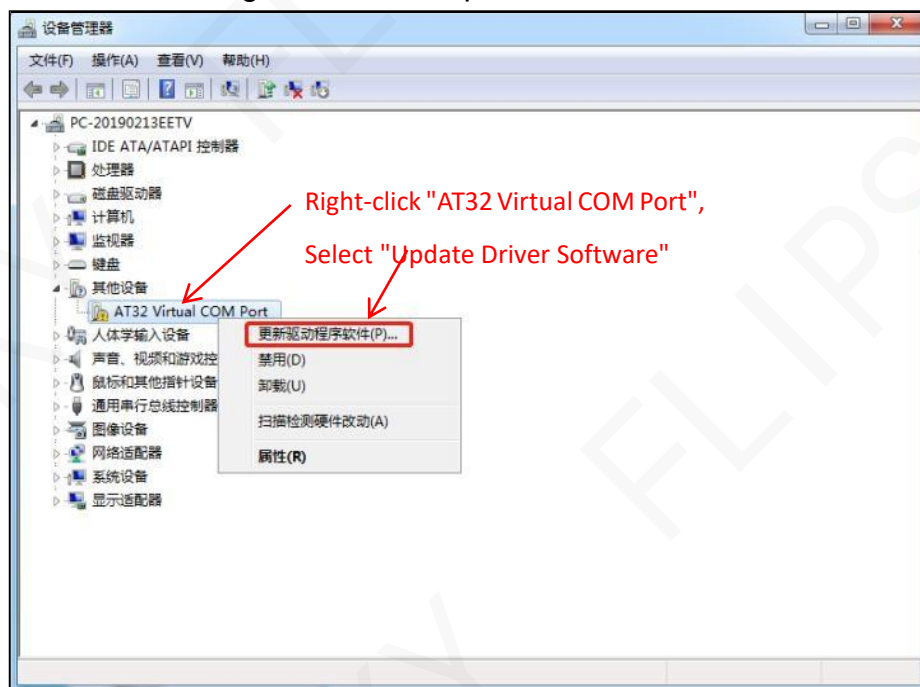
- When AT32 Virtual COM Port with a yellow icon shows as below(Fig.2-4), it means that you need to update the driver:

Screenshot 2-4. COM port driver cannot be used with a yellow icon



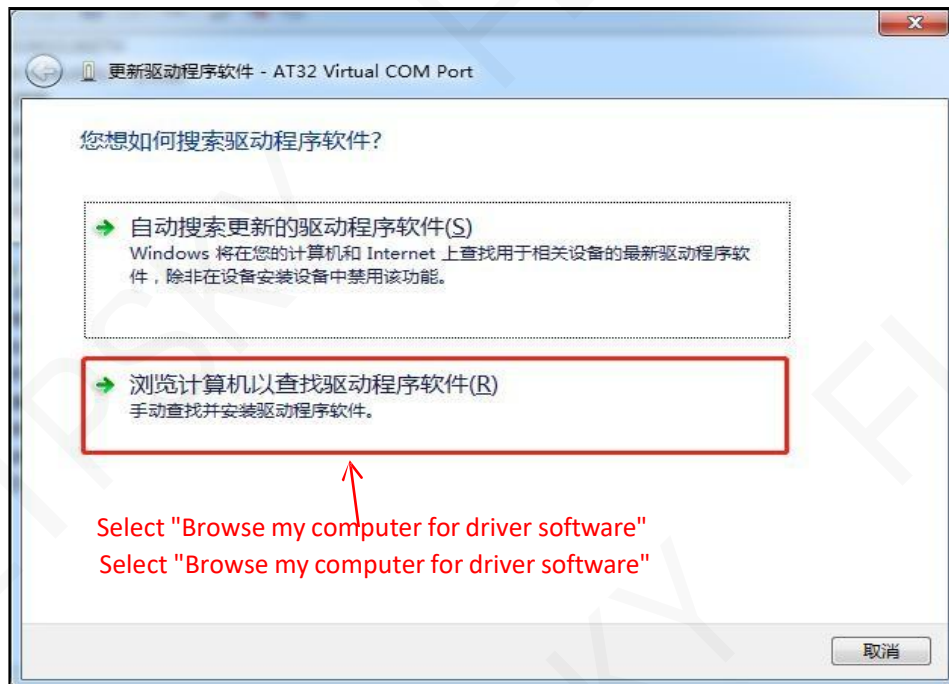
- Right-click the "AT32 Virtual COM Port" and select "Update Driver Software" as Fig.2-5.

Fig.2-5. Select "Update Driver Software"



- Select "Browse my computer for driver software" as Fig.2-6:

Fig. 2-6. Select "Browse my computer for driver software"



- Click "Browse" to find the driver file as Screenshot 2-7 below .

Screenshot 2-7. Click "Browse" to find the location driver file



- Select right driver installation folder(the folder downloaded from the link above). Take Fig.2-8 as an example:

Screenshot 2-8. Select the driver installation folder



Find and Select "Win7_or_ Win8 CDC Driver" Folder, and Click "Next" .

- If a security warning appears, select "Always install this driver software" as Fig.2-9 below:

Screenshot 2-9. Select "Always install this driver software"



Select "Always install this driver software"

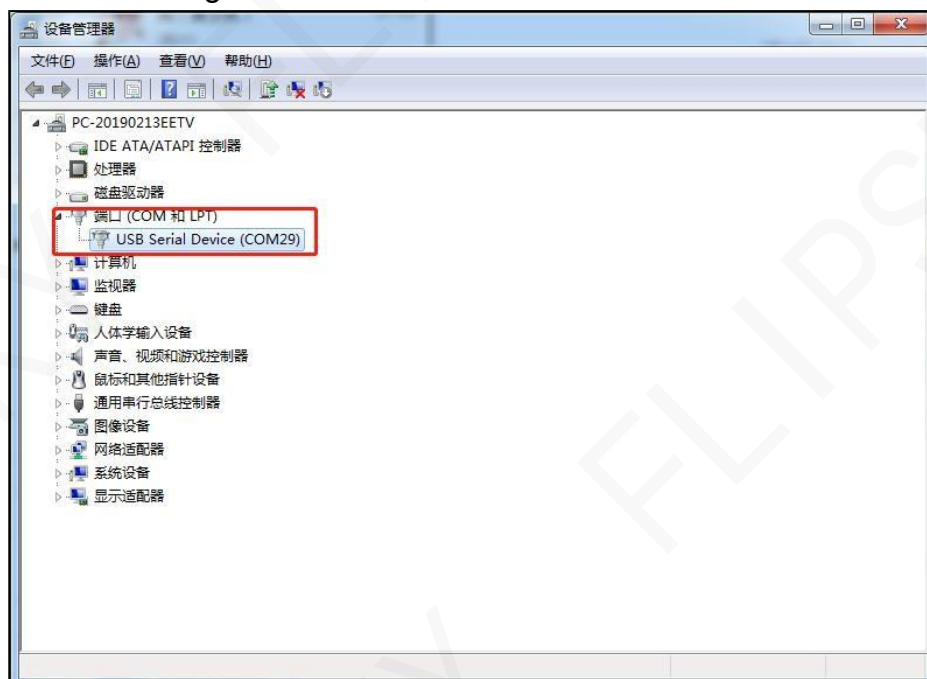
- The driver update finishes as Fig.2-10:

Fig.2-10. The Driver Installation Finish



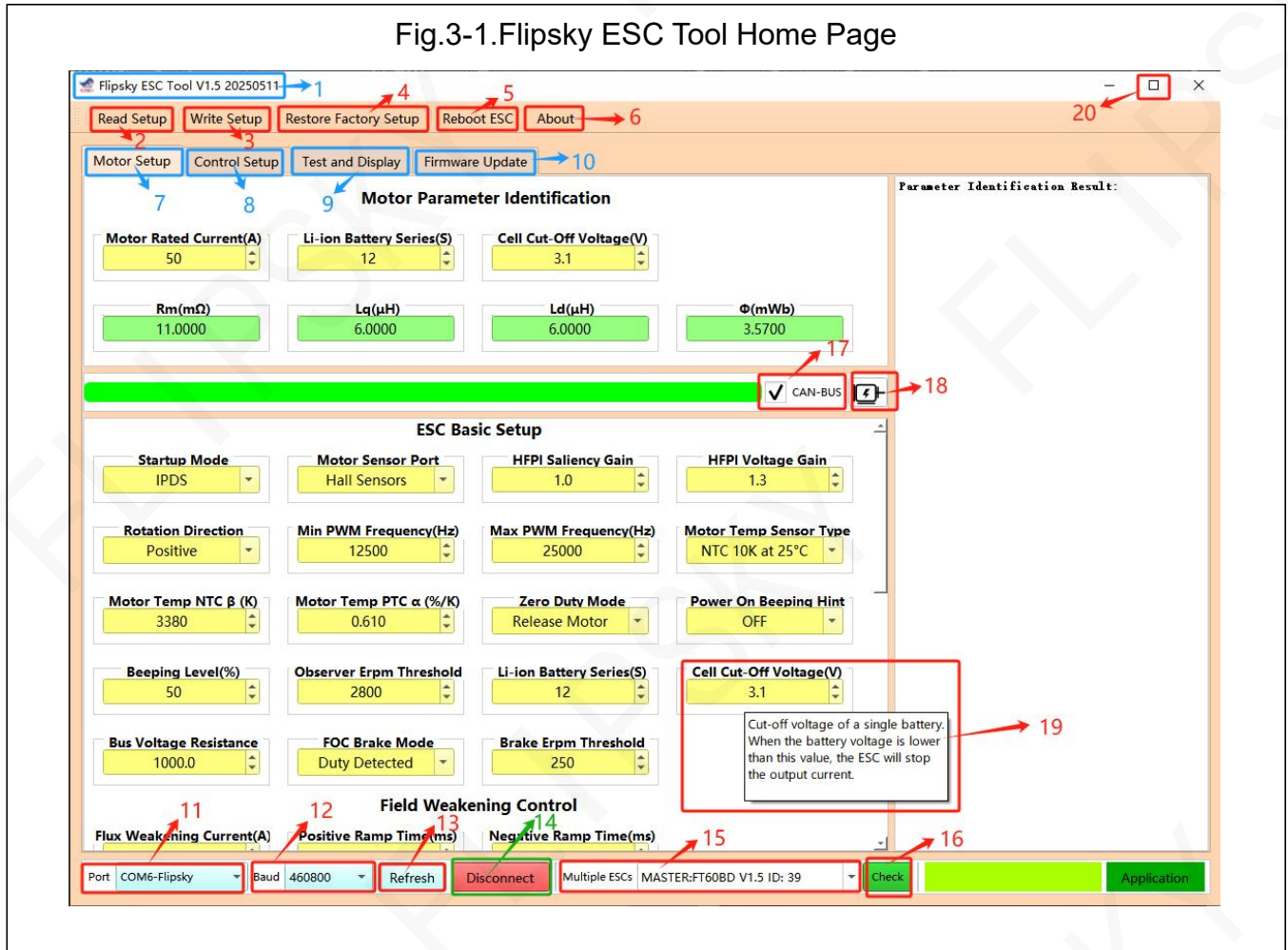
- Enter the "Device Manager" and check whether yellow icon of the USB Serial Device disappears. If so, it means that the driver has been successfully updated as Fig.2-11:

Fig.2-11. The USB Serial Device No Yellow Icon



3 Flipsky ESC Tool Home Page Introduction

Flipsky ESC Tool home page related controls introduction as Fig.3-1:



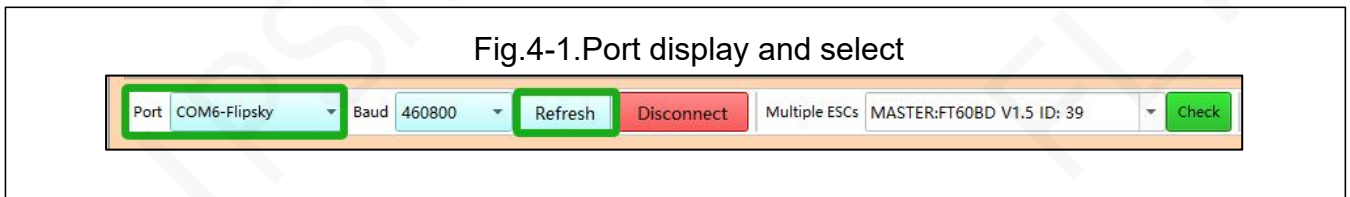
- (1) Flipsky ESC Tool V1.5 Version
- (2) Read Setup: Read the current parameters of ESC
- (3) Write Setup: Write and save the current page parameters
- (4) Restore Factory Setup: Restore factory default setup, recover all ESC parameters to the initial default values;
- (5) Reboot ESC: Click to re-connect the ESC
- (6) About: Click to read the firmware version
- (7) Motor Setup: Click to enter the motor FOC setup
- (8) Control Setup: Click to enter the input control type setup
- (9) Test and Display: Click to enter the control test and data display interface
- (10) Firmware Update: Click to enter the firmware update interface
- (11) Port: Select COM

- (12) Baud: Select baud rate
- (13) Refresh: Detect and refresh Port
- (14) Connect / Disconnect: Connect/disconnect the ESC with the ESC Tool
- (15) Multiple ESCs: Check the model, number of ESC and Master and Slave ESC ID
- (16) Check: Scan all ESCs on CAN bus
- (17) CAN-BUS: Click to identify multiple motor devices
- (18) Parameters Identification: Click to identify parameters
- (19) Parameters Definition: Explanation will be displayed when the mouse docks the parameters.
- (20) Place the mouse next to the parameters,
- (21) Maximize the window

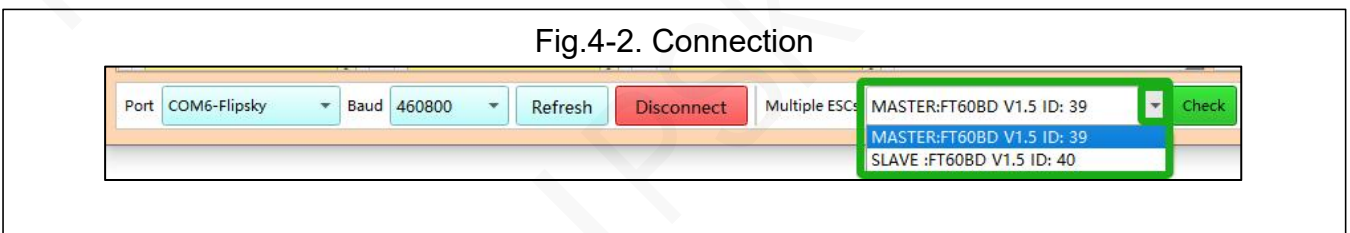
4 Build Communication Connection

4.1 Flipsky ESC Tool and FT ESC Connection

Power on the ESC and connect it with the computer via USB. Open the Flipsky ESC Tool V1.5 and the port for ESC ID (COM6-Flipsky) will be displayed. If no ID displays, please click "Refresh" to scan again as Fig.4-1.



Click "Connect" for ESC and ESC Tool connection. The ESC ID of corresponding model will appear on the screen as Fig.4-2.



The USB connection ESC is the master board, and all others connected via CAN bus are slave boards. Theoretically, 1 master board can be connected with a maximum of 253 slave CAN devices (except for master ESC ID, ESC ID=0 and ESC ID=255). While, it is recommended not using more than 8 slave boards in practice since too many CAN devices affect the working performance efficiency. Moreover, please remove the CAN terminal resistors except for the first and last ESCs' CANs. Set 1pcs Master ESC and 6pcs Slave ESCs as an example:

Master(Start)==Slave1==Slave2==Slave3==Slave4==Slave5==Slave6(End)

Remove the CAN terminal resistors of Slave1, Slave2, Slave3, Slave4 and Slave5, leaving only the CAN terminal resistors of Master (Start) and Slave6 (End).

4.2 CAN Devices Connection Error and Solution

If multiple ESCs are connected via CAN but one cannot communicate and the Flipsky ESC Tool detects that the number of ESC is less than actually connected, 3 possible causes are as below:

- 1) The slave ESC boards' CAN terminal resistors haven't been removed.
- 2) Different ESCs with same IDs
- 3) One of ESC's CAN baud rate is inconsistent with others.

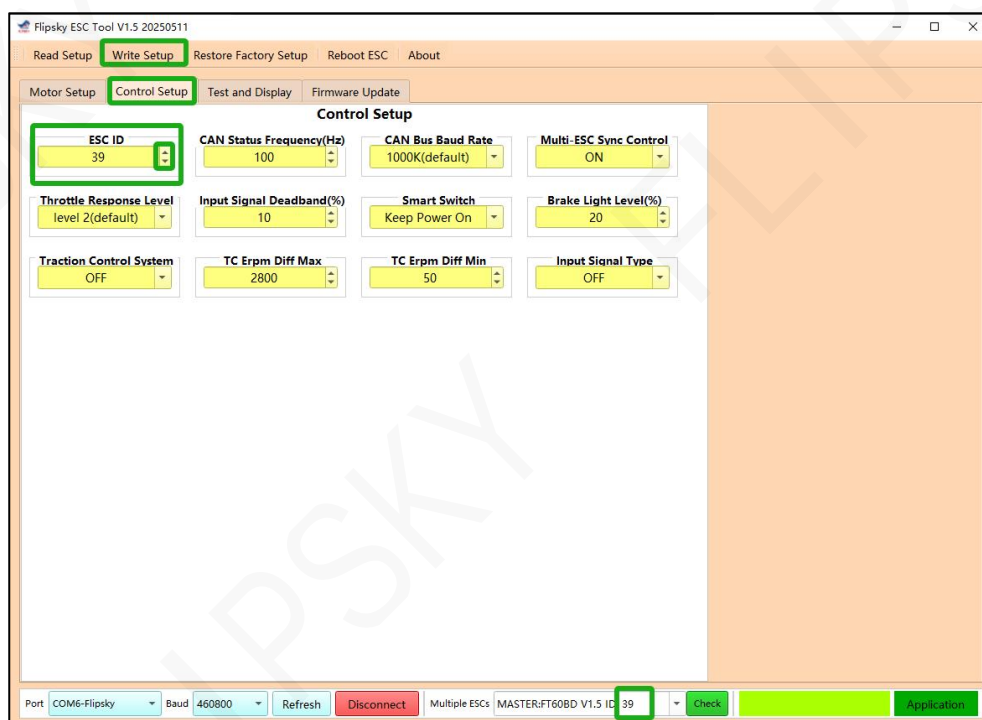
Solution:

- (1) Ensure that all CAN resistance are properly configured.(see Section 4.1)
- (2) After step (1), unplug the CAN-BUS wires. Check the ESC ID one by one by connecting them with the ESC Tool separately via USB cable. If same IDs happen, revise them to make sure the uniqueness of the ESC ID.
- (3) If it has been verified that all ESCs have different ESC IDs and the CAN end resistance has been configured correctly, but you are still unable to detect the other ESCs on the CAN bus, you can continue to check whether the CAN baud rate setting is consistent with the others. Only you can communicate normally when you make sure that the CAN baud rate setup of all ESCs are the same.
- (4) If still not succeed after the above trials, please restore all the ESCs default setup and try to reconnect.

4.2.1 ESC ID Setup

Setup steps are as Fig.4-3:

Fig.4-3.ESC ID setup



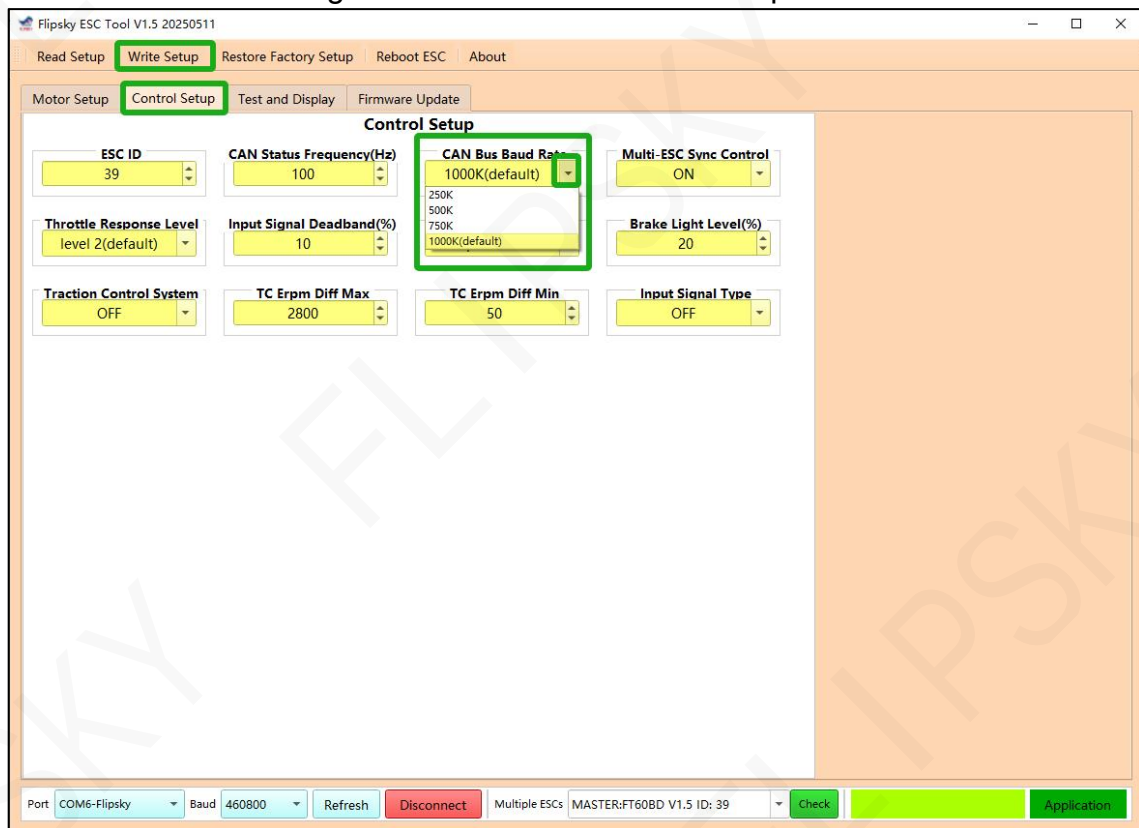
- Click "Control Setup" to enter the control setup interface
- Check the ESC ID, available range:1~254 (view details in section 7.1)
- Click "Write Setup" on the upper left to write the ESC ID into setup.

4.2.2 Set CAN Baud Rate

Setting steps are as Fig.4-4:

- Click "Control Setup" to enter the control setup interface.
- Check the CAN Bus Baud Rate in"CAN Bus Baud Rate"(view details in Section 7.2)
- Click "Write Setup" on the upper left to save the setup(as Fig.4-4)

Fig.4-4.CAN Bus Baud Rate setup



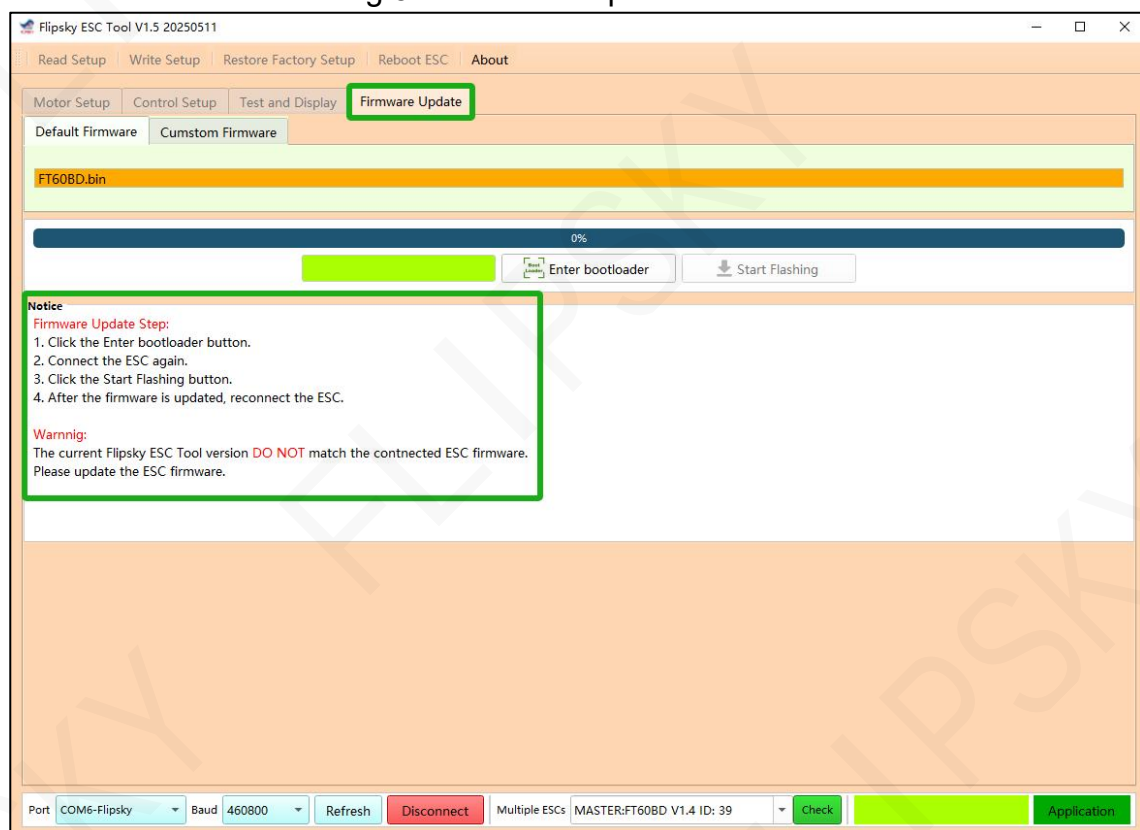
5 Firmware Update

5.1 FT Series ESC Firmware

Click “Firmware Update” to enter firmware update interface. It will automatically check if the ESC firmware version matches with the Flipsky ESC Tool. If not, please update the firmware as Fig.5-1.

NOTE: When update firmware of dual ESC(like FT85BD), please update via USB cable of the master and slave independently.

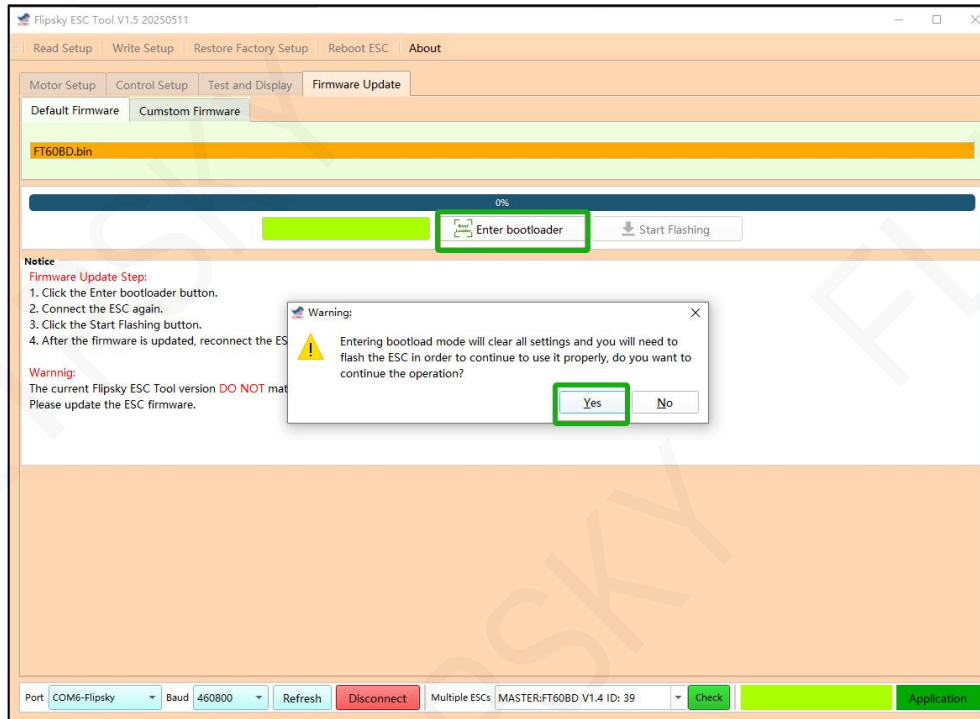
Fig.5-1.Firmware update interface



5.2 Firmware Update Steps

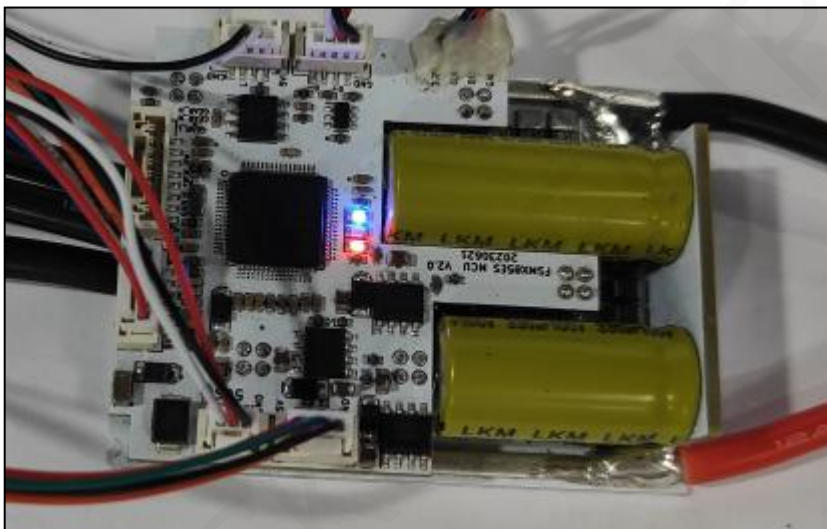
- Step 1, click “Enter bootloader” and “Yes” to enter the Bootloader mode as Fig.5-2:

P5-2.Enter bootloader

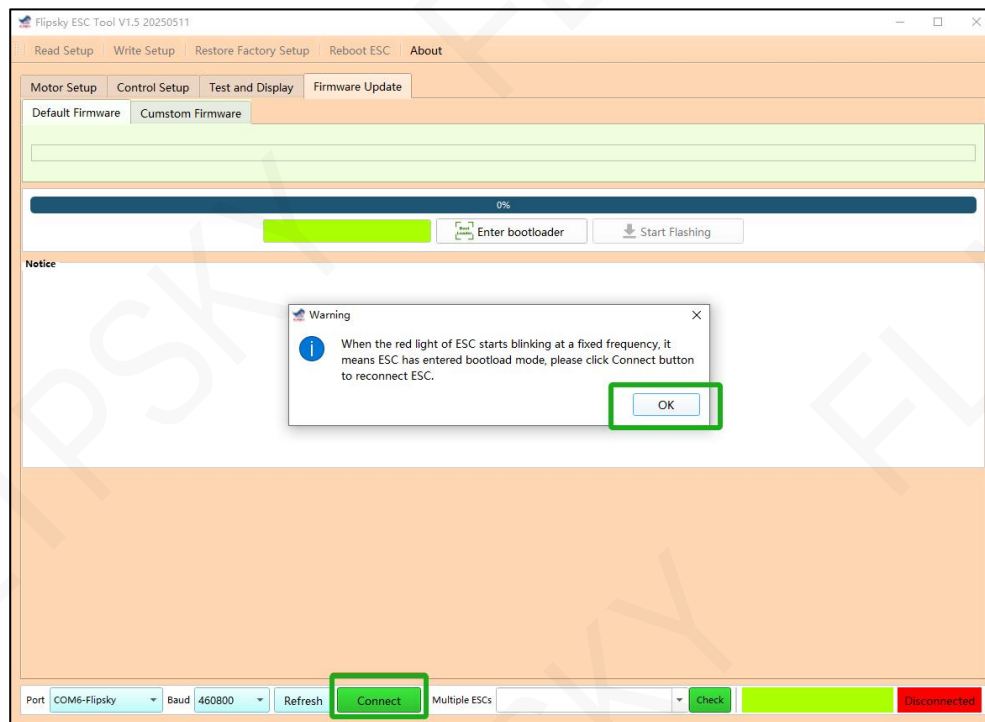


- Step 2: When the ESC red LED blinking at a fixed frequency, it means it has entered the Bootloader mode successfully (as Fig.5-3). Click "OK" and "Connect" to reconnect the ESC as Fig.5-4.

Fig.5-3.ESC enters the bootloader mode



P 5-4.Reconnect ESC



- Step 3: Click "Start Flashing", "Yes" to update corresponding firmware, as Fig.5-5, 5-6 below:

Fig.5-5.Start to update and download firmware

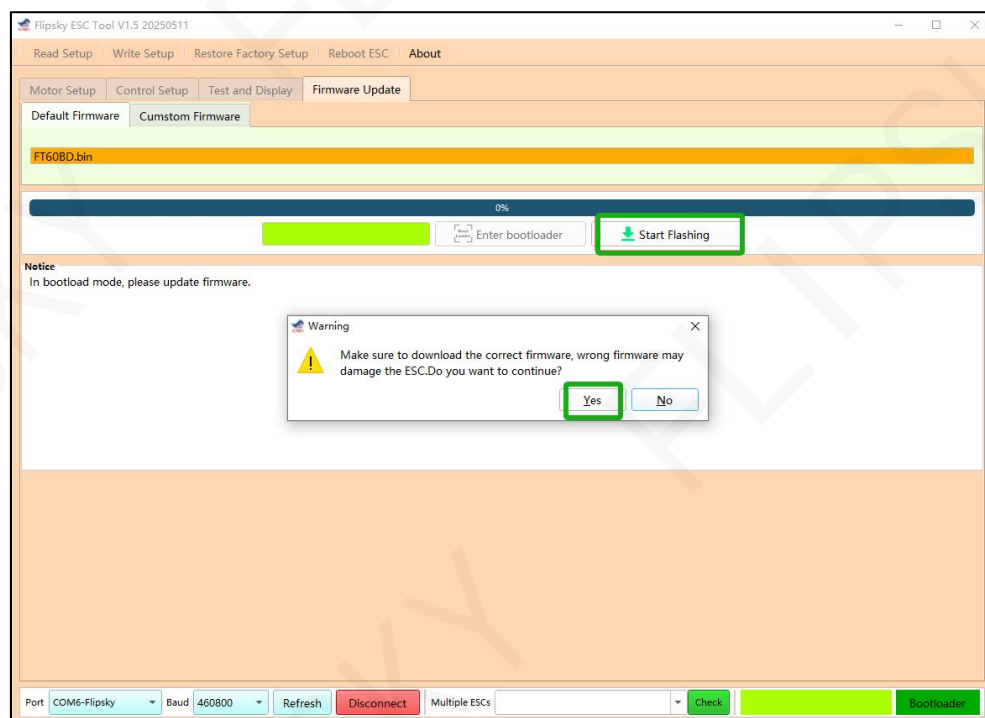
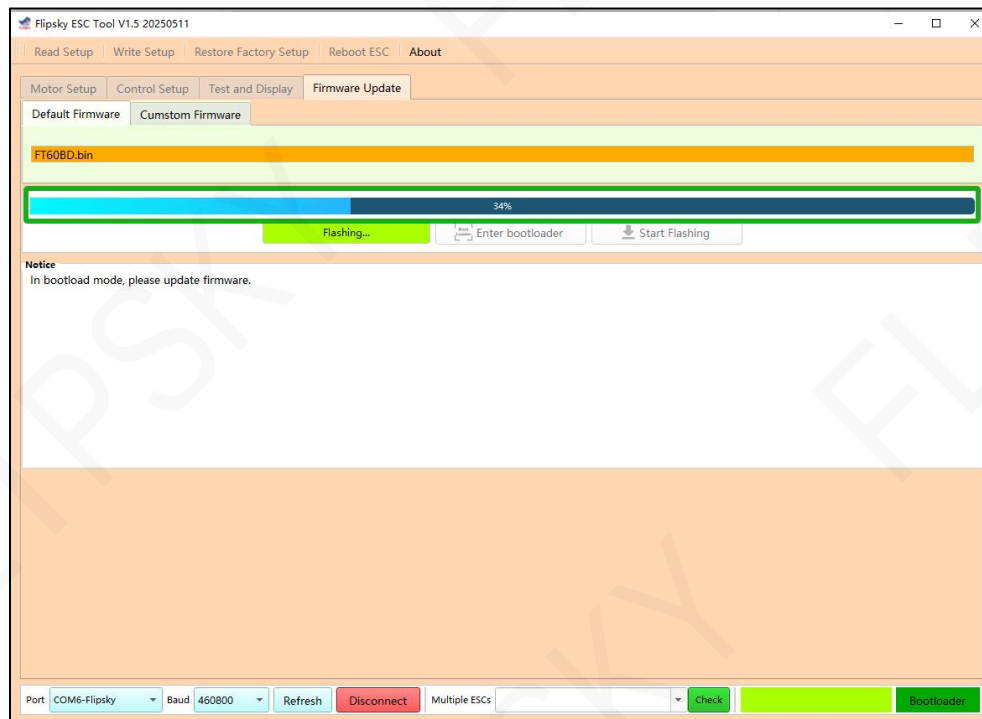
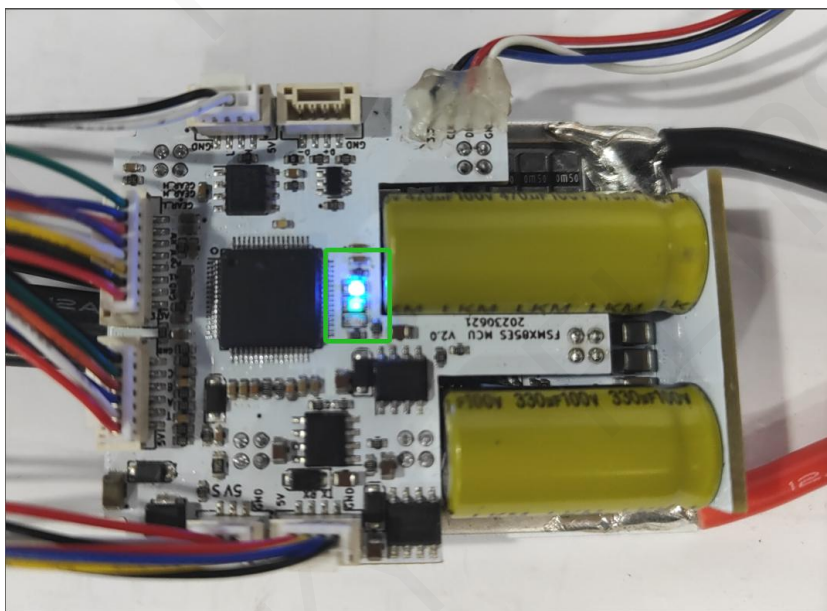


Fig.5-6.Downloading firmware



- When the ESC's green and blue LEDs keep on after the blinking of red and green LEDs, it means the firmware update succeeds(as Fig.5-7). Click "Connect" to reconnect the ESC.

Fig.5-7.Blue and green LEDs keep on



6 Motor Setup

Motor setup interface is as Fig.6-1:

Fig.6-1. Motor setup

The screenshot displays the 'Motor Setup' interface of the Flipsky ESC Tool V1.5. The window title is 'Flipsky ESC Tool V1.5 20250511'. The top menu bar includes 'Read Setup', 'Write Setup', 'Restore Factory Setup', 'Reboot ESC', and 'About'. The 'Motor Setup' tab is selected, with other tabs being 'Control Setup', 'Test and Display', and 'Firmware Update'.

The main content area is divided into several sections:

- Motor Parameter Identification:** This section contains eight parameter fields with numerical values: Motor Rated Current(A) is 50, Li-ion Battery Series(S) is 12, Cell Cut-Off Voltage(V) is 3.1, Rm(mΩ) is 11.0000, Lq(μH) is 6.0000, Ld(μH) is 6.0000, and Φ(mWb) is 3.5700.
- ESC Basic Setup:** This section contains twelve parameter fields: Startup Mode (IPDS), Motor Sensor Port (Hall Sensors), HFPI Saliency Gain (1.0), HFPI Voltage Gain (1.3), Rotation Direction (Positive), Min PWM Frequency(Hz) (12500), Max PWM Frequency(Hz) (25000), Motor Temp Sensor Type (NTC 10K at 25°C), Motor Temp NTC β (K) (3380), Motor Temp PTC α (%/K) (0.610), Zero Duty Mode (Release Motor), Power On Beeping Hint (OFF), Beeping Level(%) (50), Observer Erpm Threshold (2800), Li-ion Battery Series(S) (12), Cell Cut-Off Voltage(V) (3.1), Bus Voltage Resistance (1000.0), FOC Brake Mode (Duty Detected), and Brake Erpm Threshold (250).
- Field Weakening Control:** This section contains three parameter fields: Flux Weakening Current(A), Positive Ramp Time(ms), and Negative Ramp Time(ms).

At the bottom of the window, there is a status bar with the following information: Port (COM6-Flipsky), Baud (460800), Refresh button, Disconnect button, Multiple ESCs, MASTER:FT60BD V1.5 ID: 39, Check button, and Application button.

6.1 Motor Parameters Identification

6.1.1 Preparations for Parameters Identification

Set "Motor Rated Current(A)", "Li-ion Battery Series(S)" and "Cell Cut-Off Voltage" before doing parameters identification as below Fig.6-2.

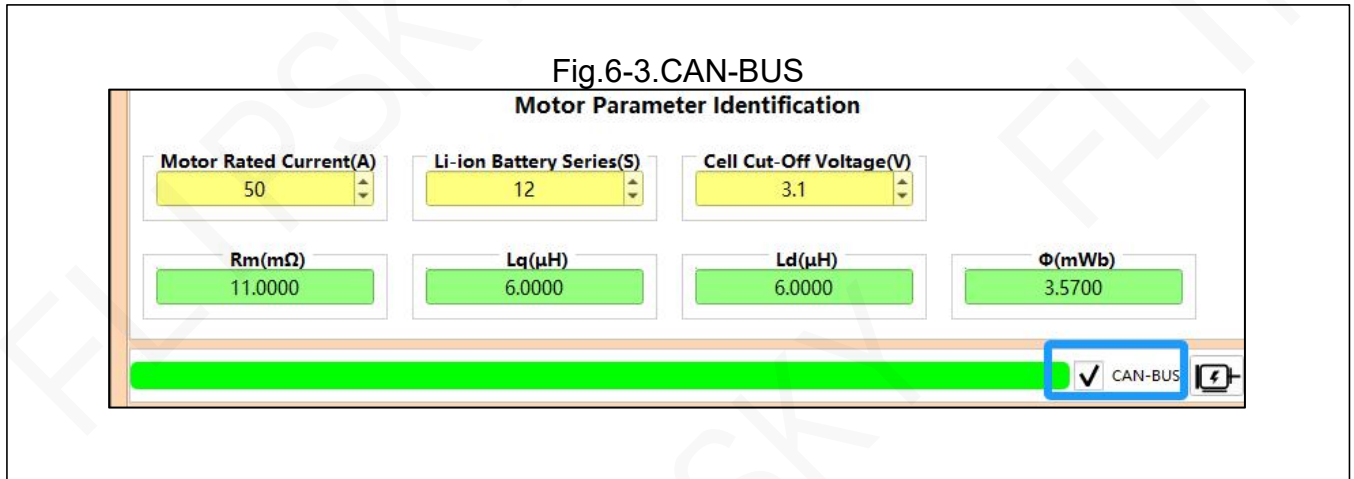
Fig.6-2.Three parameters setup before identification

The screenshot displays the 'Flipsky ESC Tool V1.5' software interface. The 'Motor Parameter Identification' section is highlighted with a blue box. It contains three dropdown menus: 'Motor Rated Current(A)' set to 50, 'Li-ion Battery Series(S)' set to 12, and 'Cell Cut-Off Voltage(V)' set to 3.1. Below these are four input fields: 'Rm(mΩ)' (11.0000), 'Lq(μH)' (6.0000), 'Ld(μH)' (6.0000), and 'Φ(mWb)' (3.5700). The 'ESC Basic Setup' section is also visible, containing various other parameters like Startup Mode, Motor Sensor Port, HFPI Saliency Gain, etc. The bottom of the window shows connection settings like Port (COM6-Flipsky), Baud (460800), and buttons for Refresh, Disconnect, Check, and Application.

- **Motor Rated Current(A):** The maximum current of motor parameters identification(A). Please adjust the current according to the motor rated current. Default as 50A, available range: 5~500A.
- **Li-ion Battery Series(S):** Write the battery Lipos per battery specifications (1S=4.2V default: 12S; adjustable range: 3-30S)
- **Cell Cut-Off Voltage(V):** Battery voltage cutoff per cell setup to prevent the battery over discharge (default: 3.1V; adjustable range: 2.6 ~ 3.3V).

6.1.2 Parameter Identification Steps

- Step 1: Set "Motor Rated Current(A)", "Li-ion Battery Series(S)", "Cell Cut-Off Voltage(V)" accordingly.(as Fig.6-2)
- Step 2: When identifying 2 motors or above, please select "CAN-BUS"(if not, don't select) as Fig.6-3 as below.



- Step 3: Click the parameter identification button(as Fig.6-4). The APP will show a warning dialog of precaution for identifying motor parameters. Click "Yes" to enter identification after confirmation.(as Fig.6-5)

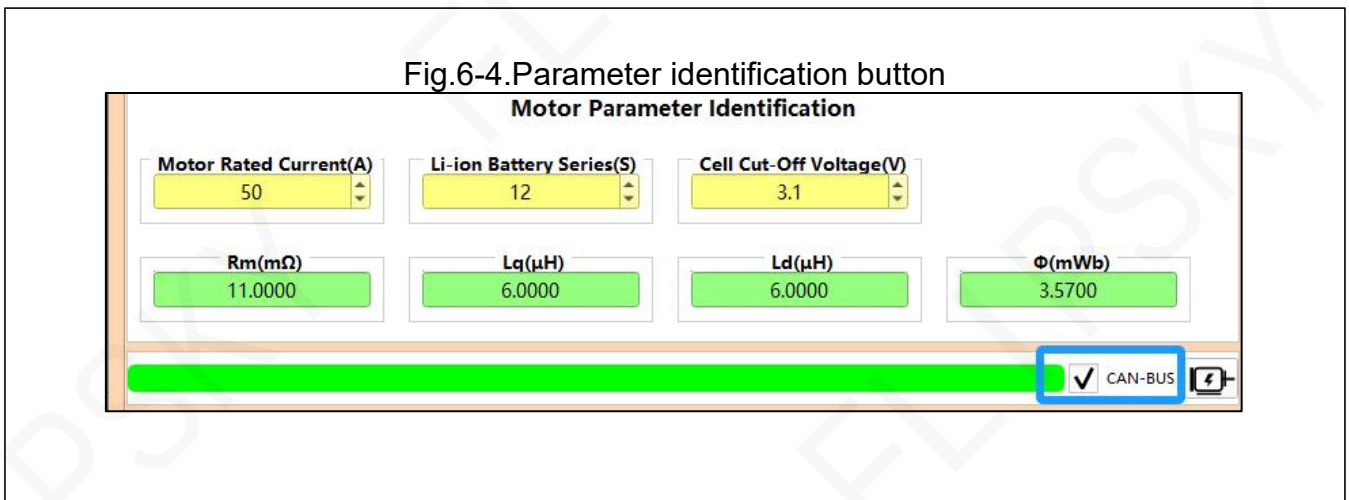
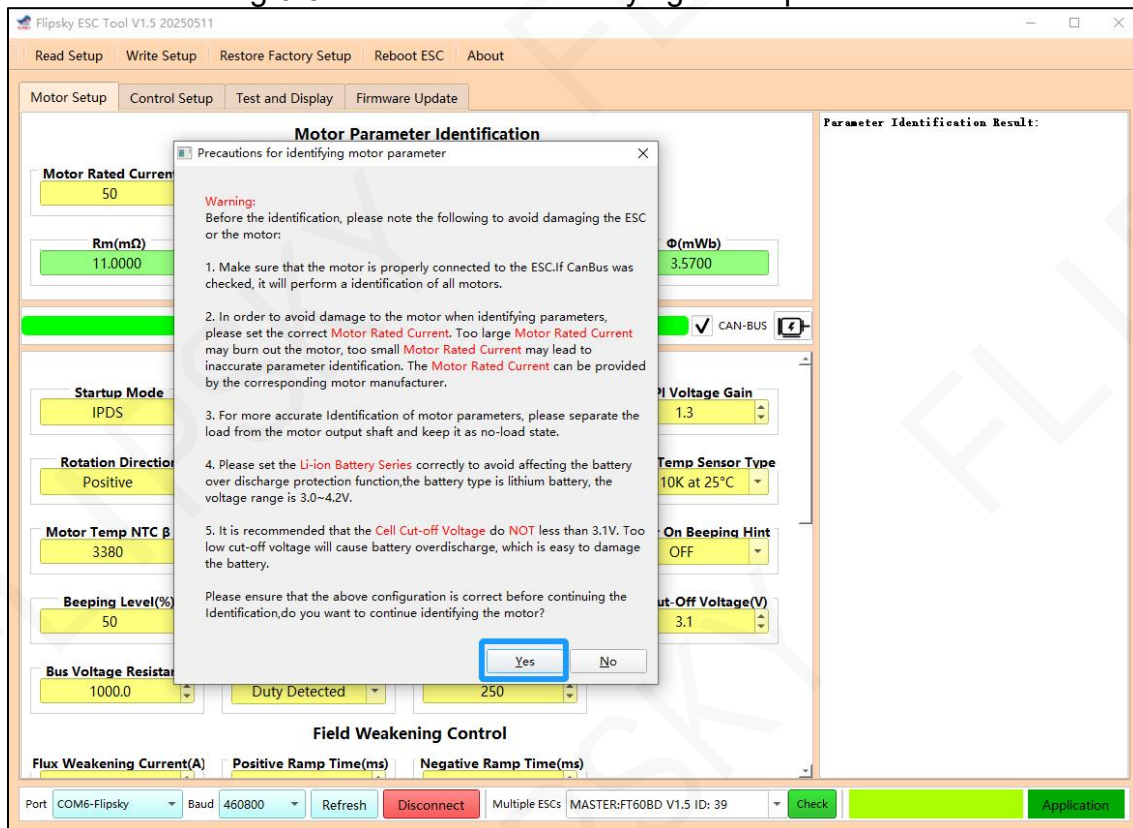


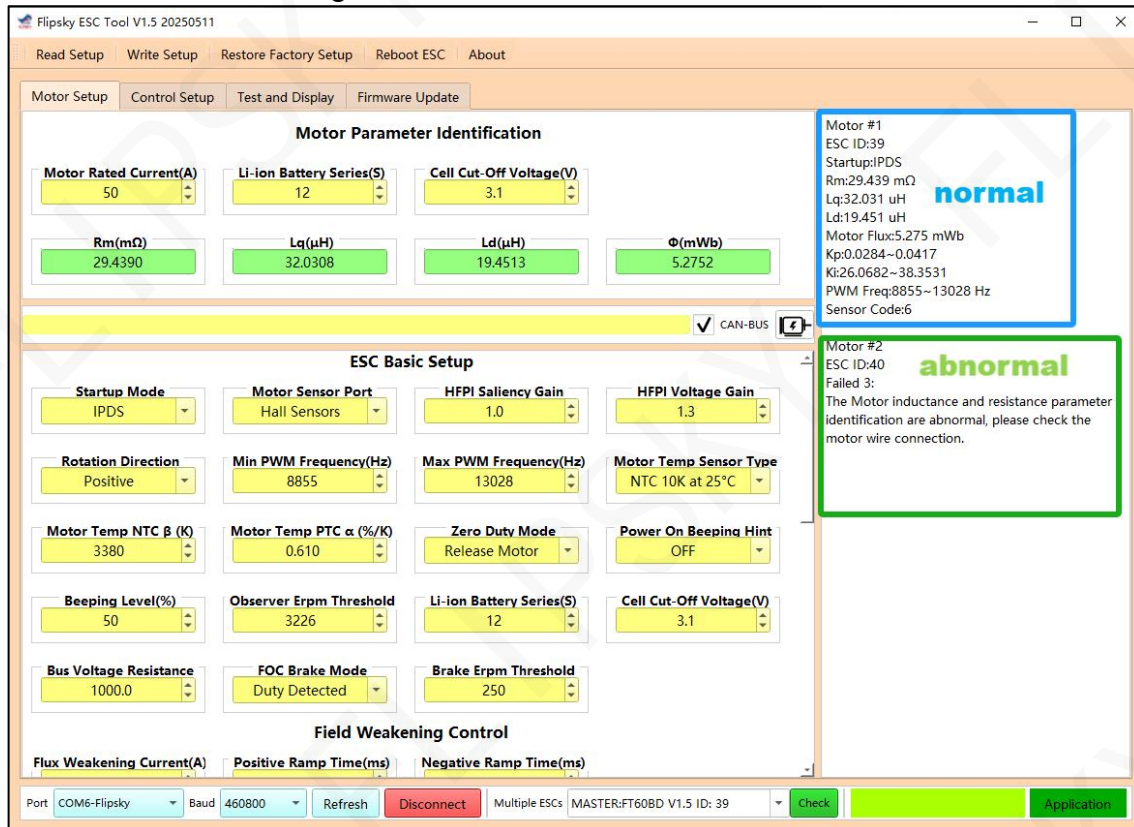
Fig.6-5.Precaution for identifying motor parameter



6.1.3 Motor Parameter Identification Results and Fault Codes

After identification, check the results on the right side of the home page.(as Fig.6-6) Check the result explanation to judge whether the identification is normal. If normal then you can continue the next step; if abnormal, find the causes through explanation.

Fig.6-6 Parameter identification results



- Motor Parameter Identification Results
- ESC ID: ESC ID
- Startup: Motor Startup Mode:
 - ✧ IPDS: Initial Position Detection Startup(sensorless mode)
 - ✧ HFPI: High Frequency Pulse Injection(sensorless mode)
 - ✧ HALL: Hall Sensor Mode
 - ✧ ENCODER: Encoder Mode
 - ✧ OPEN: OPEN Mode
- Rm: Motor Phase Resistance (mΩ)
- Lq: Motor Q-Axis Inductance (uH)
- Ld: Motor D-Axis Inductance (uH)

- Motor Flux(Φ): Motor Flux Linkage (mWb)
- Kp、Ki: Dynamic variation range of Proportional(Kp) and Integral(Ki)
- PWM Freq: Dynamic variation range of PWM Frequency
- Hall Code:
 - ✧ HALL Sensor→Hall Code: 1
 - ✧ One Hall Phase Absence→Hall Code: 2
 - ✧ Two Hall Phase Absence→Hall Code: 3
 - ✧ Hall Three-Phase Short Circuit→Hall Code: 4
 - ✧ Overloading, Undetected HALL→Hall Code: 5
 - ✧ No HALL Sensor Connection→Hall Code: 6
- **Interpretation of motor parameters identification error:**
 - Failed 1: The battery S number is set incorrectly, and does not match the actual operating voltage. Please set the correct number of battery S.

NOTE: The battery S number shall match with the actual number, or it cannot be identified/activate the over discharged protection, affecting battery lifespan.
 - Failed 2: The ESC is still faulty, please check the error code.
 - Failed 3: The Motor inductance and resistance parameter identification are abnormal, please check the motor wire connection.
 - Failed 4: The motor hall sensor identify failed. please check the motor hall sensor.
 - Failed 5: The motor flux linkage identify failed. Maybe the motor load is too heavy.

6.2 ESC Adjustable Parameters Definition and Configuration

Some motor parameters like R_m , L_q , L_d , Φ etc. will be calculated automatically after the parameter identification finish which don't need to change manually. Some control parameters (e.g. Startup Mode, HFPI Saliency Gain, HFPI Voltage Gain, Max PWM Frequency, Min PWM Frequency, etc.) will be reasonably derived by the adaptive algorithm according to the motor characteristics and whether using the HALL sensor. If not necessary, these parameters do not need to be modified. Other parameters (e.g. Max Erpm, Max Current) can be manually set accordingly.

Adjustable parameters including: ESC Basic Setup, Field Weakening Control, Max Erpm Setup, Max Current Setup, Encoder Setup, rpm PID setting and location PID setting, as Fig.6-7.

NOTE: Remember to click the upper left button "Write Setup" to save the configurations.

Fig.6-7 Adjustable Parameters

The screenshot shows the Flipsky ESC Tool V1.5 interface. The top menu bar includes 'Read Setup', 'Write Setup' (highlighted with a green box), 'Restore Factory Setup', 'Reboot ESC', and 'About'. The 'Motor Setup' tab is selected, showing a blue border around its content area. The 'Motor Parameter Identification' section contains fields for Motor Rated Current (A) set to 50, Li-ion Battery Series (S) set to 12, Cell Cut-Off Voltage (V) set to 3.1, R_m (m Ω) set to 29.4390, L_q (μ H) set to 32.0308, L_d (μ H) set to 19.4513, and Φ (mWb) set to 5.2752. The 'ESC Basic Setup' section includes Startup Mode (IPDS), Motor Sensor Port (Hall Sensors), HFPI Saliency Gain (1.0), HFPI Voltage Gain (1.3), Rotation Direction (Positive), Min PWM Frequency (Hz) (8855), Max PWM Frequency (Hz) (13028), Motor Temp Sensor Type (NTC 10K at 25°C), Motor Temp NTC β (K) (3380), Motor Temp PTC α (%/K) (0.610), Zero Duty Mode (Release Motor), Power On Beeping Hint (OFF), Beeping Level (%) (50), Observer Erpm Threshold (3226), Li-ion Battery Series (S) (12), Cell Cut-Off Voltage (V) (3.1), Bus Voltage Resistance (1000.0), FOC Brake Mode (Duty Detected), and Brake Erpm Threshold (250). The 'Field Weakening Control' section at the bottom includes Flux Weakening Current (A), Positive Ramp Time (ms), and Negative Ramp Time (ms). On the right, a panel displays Motor #1 details: ESC ID:39, Startup:IPDS, R_m :29.439 m Ω , L_q :32.031 μ H, L_d :19.451 μ H, Motor Flux:5.275 mWb, K_p :0.0284~0.0417, K_i :26.0682~38.3531, PWM Freq:8855~13028 Hz, and Sensor Code:6. Below this, Motor #2 details show ESC ID:40 and a 'Failed 3:' error message: 'The Motor inductance and resistance parameter identification are abnormal, please check the motor wire connection.' The bottom status bar shows 'Port COM6-Flipsky', 'Baud 460800', 'Refresh', 'Disconnect', 'Multiple ESCs', 'MASTER:FT60BD V1.5 ID: 39', 'Check', and 'Application' buttons.

6.2.1 Basic ESC Parameters Setup

Basic ESC parameters setup is as Fig.6-8:

Fig.6-8.Basic ESC parameters

Startup Mode IPDS	Motor Sensor Port Hall Sensors	HFPI Saliency Gain 1.0	HFPI Voltage Gain 1.3
Rotation Direction Positive	Min PWM Frequency(Hz) 8686	Max PWM Frequency(Hz) 12916	Motor Temp Sensor Type NTC 10K at 25°C
Motor Temp NTC β (K) 3380	Motor Temp PTC α (%/K) 0.610	Zero Duty Mode Release Motor	Power On Beeping Hint OFF
Beeping Level(%) 50	Observer Erpm Threshold 3228	Li-ion Battery Series(S) 12	Cell Cut-Off Voltage(V) 3.1
Bus Voltage Resistance 1000.0	FOC Brake Mode Duty Detected	Brake Erpm Threshold 250	

● Startup Mode:

Three motor startup modes can be selected:

- ✧ IPDS: Initial Position Detection Startup (suitable for light loading sensorless mode such as underwater DPVs, multi-axis FPVs etc.)
- ✧ HFPI: High Frequency Pulse Injection (suitable for heavy loading sensorless mode such as E-Bike, E-Board, mini buggy etc.)
- ✧ HALL: Hall Sensor Startup(widely used, especially suitable for equipment with full load)
- ✧ ENCODER: Encoder Sensor Startup
- ✧ OPEN: Open Mode

NOTE: HFPI mode is not suitable for all motors. If there are serious vibrations, stalls etc. when using this mode, it means that this mode is not suitable for the motor. If the motor integrates HALL interface, we strongly recommend using HALL for smooth running and better performance.

● Motor sensor port:

Before setup, please choose the corresponding motor sensor port and remember to click "Write Setup".

- ✧ Hall sensor: Three Hall effect position sensors. A Hall effect component is installed every 60°/120° electrical Angle.
- ✧ ABI Encoder: An incremental encoder is a linear or rotary electric-mechanical device that has two output signals, A and B, which issue pulses when the device is moved.

Note: Since the incremental encoder cannot determine the initial position of the rotor, it is recommended to rotate the motor manually one circle for zero position calibration after the first start of the ESC, otherwise unexpected situations may occur when the control command is forcibly sent.

- ✧ AS5047(SPI): An absolute encoder maintains position information when power is removed from the encoder, which will immediately report changes and no need to do zero position calibration.

● HFPI Saliency Gain

Adjustable range: 0.5~4.0

Parameter Specification: The surface-mounted permanent magnet synchronous motor, because of its weak convex polarity, has a large noise in the angle measurement when using the HFPI mode, and this parameter is designed to improve the accuracy of the angle measurement. This value is automatically done by the algorithm after the motor parameters are recognized. The algorithm will calculate a reasonable value according to the identification results. **It's no need to change if not necessary.**

● HFPI Voltage Gain

Adjustable range: 0.2~3.0

Parameter Specification: Parameter description: This parameter setup is designed for high-frequency pulse voltage injection needs of motor D-axis in HFPI mode, which is determined by the motor specification and can be matched automatically after the motor parameter identification finish, so **it's no need to write manually if not necessary.**

● Rotation Direction

- ✧ Positive
- ✧ Negative

● Min PWM Frequency

Adjustable range: 2000~25000

Parameter Specification: This parameter can be matched automatically after the motor parameter identification finish and **it's no need to write manually if not necessary.**

● Max PWM Frequency

Adjustable range: 10000~25000

Parameter Specification: This parameter can be matched automatically after the motor parameter identification finish and **it's no need to write manually if not necessary.**

- **Motor Temp sensor Type**

- ✧ NTC 10K at 25°C
- ✧ NTC 10K at 25°C
- ✧ NTC 100K at 25°C
- ✧ PT1000PTC 1K at 100°C
- ✧ KTY83-122
- ✧ KTY83-152
- ✧ KTY84-130
- ✧ KTY84-150
- ✧ KTY84-151
- ✧ KTY84-152

Parameter Specification: NTC 10K temperature sensor with a temperature of 25°C is the most widely used one in applications but some others are also used in special application fields.

- **Motor Temp NTC β (K)**

Adjustable range: 1~10000

Parameter Specification: A thermistor's " β " value, or beta value, is an indication of the shape of the curve representing the relationship between resistance and temperature of an NTC thermistor. Please write the correct data per your motor. Flipsky's motor NTC β is 3380.

- **Motor Temp PTC α (%/K)**

Adjustable range: 0.5~100

Parameter Specification: The PTC type temperature sensor α coefficient value. Please select corresponding temperature sensor type.

- **Zero Duty Mode**

- ✧ Keep Output
- ✧ Release Motor

Parameter Specification: Zero Duty Mode. If set duty cycle 0% in Zero Duty Mode, users can choose the mode of Keep Output or Release Motor.

- **Power on Beeping Hint**

- ✧ OFF

- ✧ ON

Parameter Specification: When activate the function, the ESC will beep for 3 times after powering on.

Note: Motor parameter identification must be completed before turning on the function, otherwise the sound is abnormal.

- **Beeping Level(%)**

Adjustable range: 0~1

Parameter Specification: The larger of the value, the louder of the sound.

- **Observer Erpm Threshold**

Adjustable range: 1~250000

Parameter Specification: When the rpm is over than this value, switch to the observer erpm threshold to measure the rotor position and rpm.

- **Li-ion Battery series(s)**

Adjustable range: 3~30S;

Parameter Specification: Li-ion battery series setup. Set according to the battery voltage(1S=4.2V)

- **Cell cut-off Voltage(V)**

Adjustable range: 2.6~3.3V

Parameter Specification: Single cell battery cut-off voltage settings to prevent over-discharge of the battery and prolong the battery lifespan. Default as 3.1V

- **Bus Voltage Resistance**

Adjustable range: 900~1100

Parameter Specification: Adding motor "Bus Voltage Resistance" setup to make the power supply voltage detected more accurately (adjust according to the voltage measured by the multimeter or the power supply voltage). If not necessary, keep the default value without modification.

- **FOC Brake Mode**

Available mode: Duty Cycle Detected、Erpm Detected、Duty Erpm Detected。

Parameter Specification:

- ✧ Duty Cycle Detected: When enter brake status and detect the Duty cycle is lower than 0.001, switch to the three-phase short-circuit status.
- ✧ Erpm Detected: When enter brake status and detect the Erpm is lower than Brake Erpm Threshold, switch to the three-phase short-circuit status.
- ✧ Duty Erpm Detected: When detect the duty cycle is lower than 0.001 and the absolute value of Erpm is lower than Brake Erpm Threshold in the meanwhile, switch to the three-phase short-circuit status.

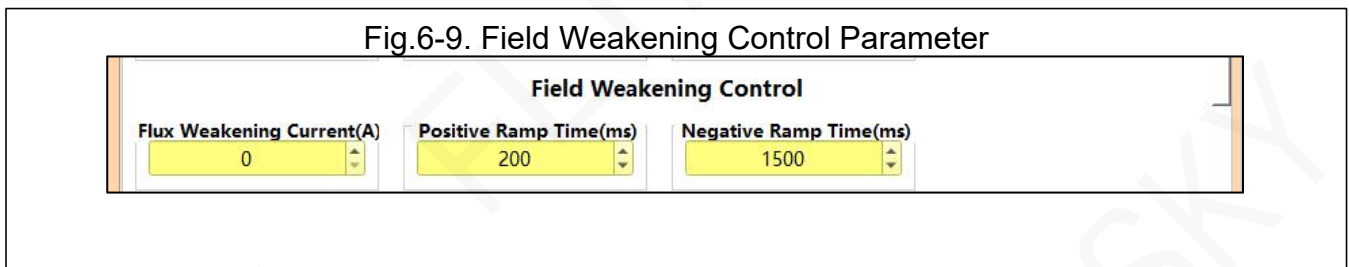
● Brake Erpm Threshold

Adjustable range: 0~8000

Parameter Specification: When in this brake control status, if the detected erpm is lower than this brake Erpm threshold, the ESC will switch to the three-phase short-circuit status.

6.2.2 Field Weakening Control Parameter Setup

Field weakening control parameters including “Flux Weakening Current(A)”、“Positive Ramp Time(ms)”、“Negative Ramp Time(ms)”；as Fig.6-9:



● Flux Weakening Current

Adjustable range: 0~500

Parameter Specification: the current applied to the D-axis in field weakening control. Increasing this value can increase the maximum speed limit of the motor. The higher the value, the higher the maximum speed, but at the same time, the higher the current consumption and the hotter the motor, it is recommended to set it lower than 1/4 of the Motor Current Max.

● Flux Weakening Pos ramp time

Adjustable range: 0~10000

Parameter Specification: When the Positive Ramp Time is set as 1000ms, the flux weakening current to 20A, the flux weakening Current will increase from 0A to 20A in 1000ms when the duty cycle

reaches 90% or above.

- **Flux Weakening Neg ramp time**

Adjustable range: 0~10000

Parameter Specification: When the Negative Ramp Time is set as 1000ms, and the current control is set to be 0, the flux weakening current will fall from 20A to 0A in 1000ms and ends the flux weakening control.

6.2.3 Max Erpm Setup

The max erpm setup includes “Positive Erpm”, “Negative Erpm”, “H-Gear Erpm”, “M-Gear Erpm”, “L-Gear Erpm”, “R-Gear Max Erpm” as Fig.6-10.

Fig.6-10.Max Erpm Setup

Max Erpm Setup

Positive Erpm 200000	Negative Erpm -200000	H-Gear Erpm 100000	M-Gear Erpm 26000
L-Gear Erpm 13000	R-Gear Erpm 4000		

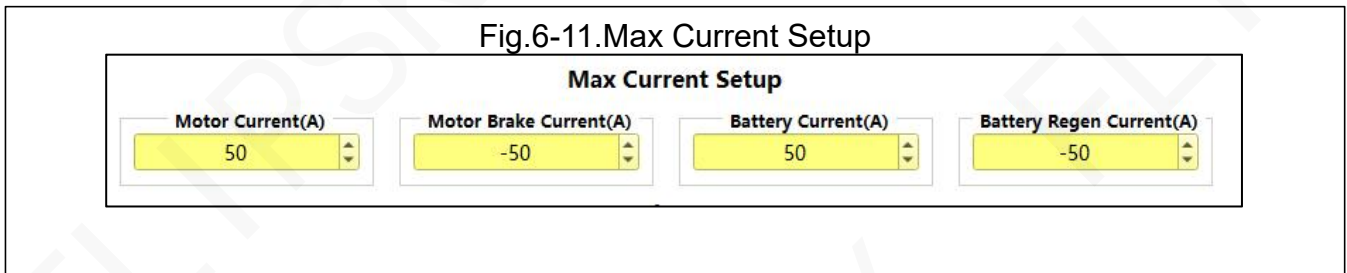
- **Positive Max Erpm:** the maximum forward erpm of motor
Adjustable range: 500~250000
- **Negative Max Erpm:** the maximum reverse erpm of motor
Adjustable range: -250000~-500
- **H-Gear Max Erpm:** the maximum erpm of motor in high speed gear
Adjustable range: 500~250000
- **M-Gear Max Erpm:** the maximum erpm of motor in medium speed gear
Adjustable range: 500~250000
- **L-Gear Max Erpm:** the maximum erpm of motor in low speed gear
Adjustable range: 500~250000

- **R-Gear Max Erpm:** the maximum erpm of motor in reverse gear

Adjustable range: 500~250000

6.2.4 Max Current Setup

The max current setup includes maximum motor current and maximum battery current setup. The maximum motor current setup includes “Motor Current” and “Motor Brake Current”. The maximum battery current setup includes “Battery Current” and “Battery Regen Current” as Fig.6-11:



NOTE: The max(motor/current) current limit of every ESC of FT series is different.

- **Max Motor Current:** the maximum continuous working current (A)

Adjustable range: 0~550

- **Max Motor Brake Current:** the maximum continuous braking current (A)

Adjustable range: -550~0

NOTE: If using the DC power supply, the Max Motor Brake Current must be set to “0”. Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed state.

- **Max Battery Current:** the maximum continuous output current(A)

Adjustable range: 0~550

NOTE: When setting Max Battery Current, it must be set according to the battery specification.

Set a battery with maximum continuous output current 50A as an example:

single ESC, Max Battery Current: 50A

dual ESC, Max Battery Current: $50/2=25A$

4WD ESC, Max Battery Current: $50/4=12.5A$

The total output current of the battery must not exceed the battery maximum continuous output current specification. Otherwise, the BMS will cut off the power supply and make the motor lose control, which is very dangerous. Moreover, please make sure that the Max Battery Current setting value does not exceed than that of the DC power supply to avoid ESC damage or other unpredictable

conditions.

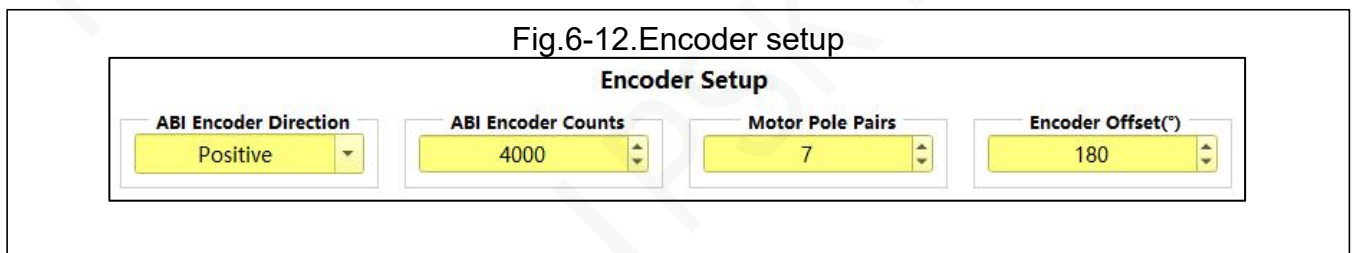
- **Max Battery Regen Current:** the maximum continuous regenerative current of battery (A)

Adjustable range: -550~0

Note: If using the DC power supply, the Max Battery Regen Current must be set to "0". Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed mode.

6.2.4 Encoder Setup

Encoder setup includes "ABI Encoder Direction", "ABI Encoder Counts", "Motor Pole Pairs", "Encoder Offset(°)" as Fig.6-12.



- **ABI Encoder Direction**

- ◇ Positive
- ◇ Negative

Parameter Specification: If the A and B phases are connected in reverse, the pulse count will be reversed. In this case, the encoder direction needs to be corrected.

- **ABI Encoder Counts**

Adjustable range: 0~100000

Parameter Specification: This parameter is 4 times the number of pulses output by the encoder when it rotates one circle. For example, if both phase A and phase B output 1000 pulses when they rotate one circle, they should be set to 4000 as 4 times of the frequency.

- **Motor Pole Pairs**

Adjustable range: 0~100

Parameter Specification: When the motor sensor port is set to encoder mode, this value will be

calculated automatically after finishing the motor parameter identification. If no special application, no modification is required. Note: When starting in encoder mode, please ensure the correct setting of this value.

- **Encoder Offset(°)**

Adjustable range: 0~360°

Parameter Specification: It means the offset angle between the encoder and the motor rotor. When the motor sensor port is set to encoder mode, this value will be calculated automatically after finishing the motor parameter identification. If no special application, no modification is required.

6.2.5 Speed PID Setup

Speed PID setup includes “Speed PID Kp”, “Speed PID Ki”, “Speed PID Kd”, “PID Minimum Erpm”, “PID Maximum Erpm”, “PID Accel ($r/(m^2)$)Erpm” as Fig.6-13:

Fig.6-13.Speed PID setup

Speed PID Setup

Speed PID Kp 0.00380	Speed PID Ki 0.00380	Speed PID Kd 0.00010	PID Minimum Erpm 500
PID Maximum Erpm 100000	PID Accel($r/(m^2)$) 30000		

- **Speed PID Ki: Speed PID controller integral gain coefficients.**

Adjustable range: 0~10000

If no special application, no modification is required.

- **Speed PID Kd: Speed PID controller differential coefficients**

Adjustable range: 0~10000

If no special application, no modification is required.

- **PID Maximum Erpm: Minimum speed for speed PID control.**

Adjustable range: 0~250000

Parameter Specification: When the ESC detects the Erpm higher than this max. value, it will maintain the maximum speed PID Maximum Erpm output.

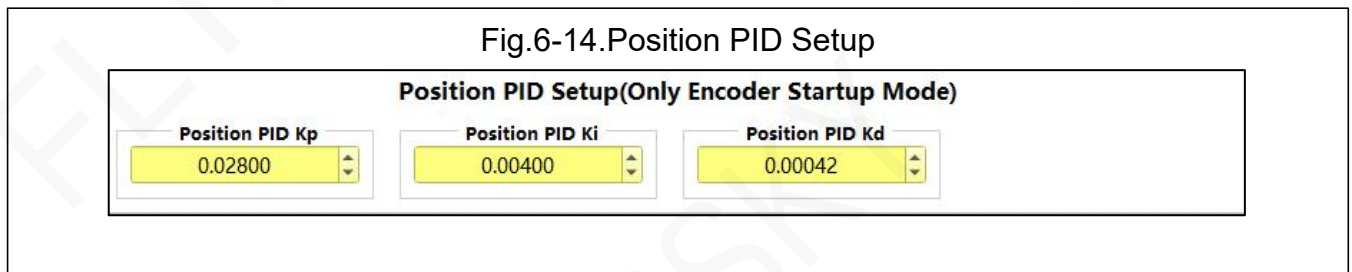
- **PID Accel ($r/(m^2)$)Erpm: Acceleration value for speed PID control.**

Adjustable range: -1~150000

Parameter Specification: The larger the value, the faster the speed changes. For example, when it is set to 30000 Erpm, it means that the speed changes by 30000 Erpm in 1 second. Which means that it takes 1 second only for the motor to increase Erpm from 0 to 30000 when a speed command of 30000 Erpm is given.

6.2.6 Position PID Setup(Encoder Startup Mode only)

Position PID Setup includes "Position PID Kp", "Position PID Ki" and "Position PID Kd" as Fig.6-14 below:



- **Position PID Kp: Position PID controller proportional gain coefficients.**

Adjustable range: 0~10000

If no special application, no modification is required.

- **Position PID Ki: Position PID controller integral gain coefficients.**

Adjustable range: 0~10000

If no special application, no modification is required.

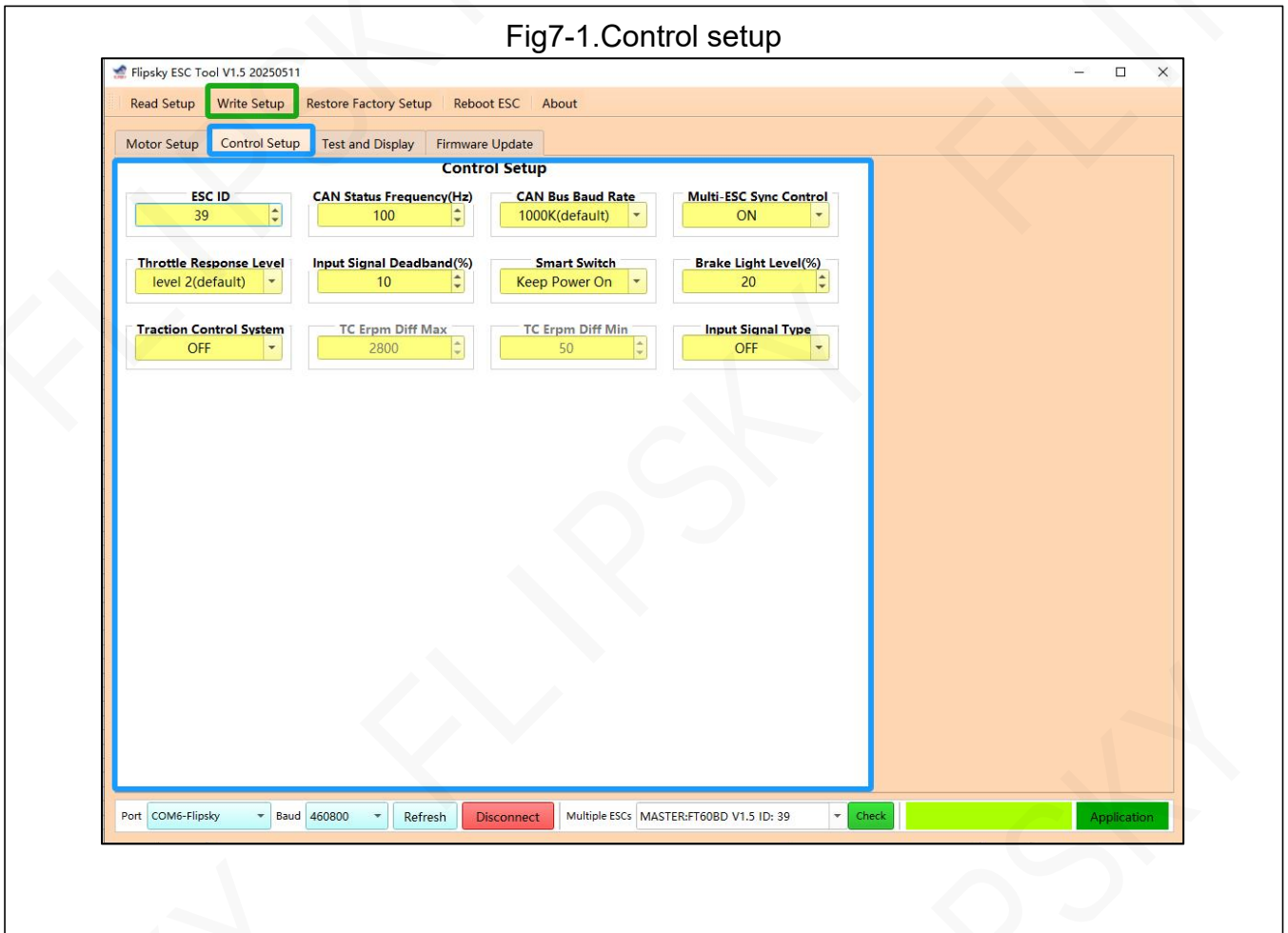
- **Position PID Kd: Position PID controller derivative coefficients.**

Adjustable range: 0~10000

If no special application, no modification is required.

7 Control Setup

Control setup parameter includes “ESC ID”, “CAN Status Frequency(Hz)”, “CAN Bus Baud Rate”, “Multi-ESC Sync Control”, “Throttle Accel Level”, “Input Signal Deadband”, “Smart Switch”, “Brake Light Level(%)”, “Traction Control Svstem”, “TC Erpm Diff Max”, “TC Erpm Diff Min” and “Input Signal Type” as Fig.7-1 as below:



Setup Steps are as follow:

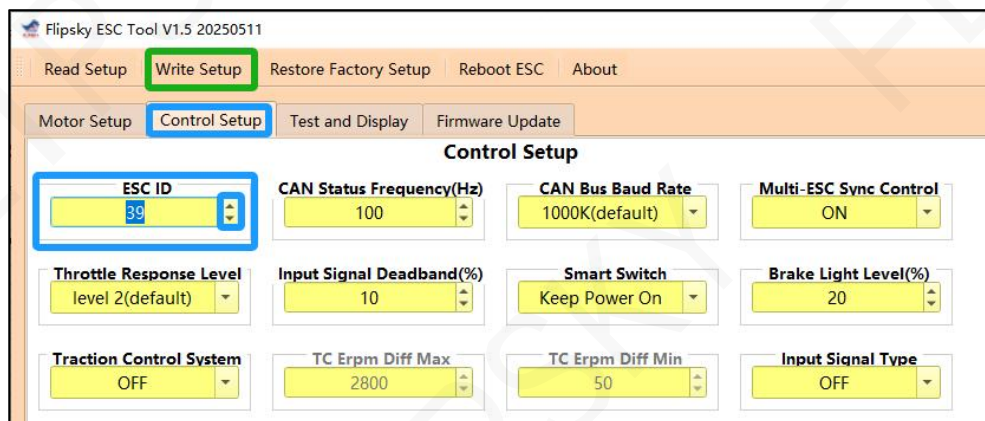
- Click “Control Setup” to enter the control setup interface
- Set the needed parameters
- Click “Write Setup” on the upper left to save the configuration

7.1 ESC ID Setup

ESC ID: ESC ID(Fig.7-2), adjustable range: 1~254. For distinguish the ESCs on the same CAN bus. Please make sure the uniqueness of the ESC ID since the same ESC IDs result in parameter identification failures or abnormal control. ESC ID 0 (special command number) and ESC ID 255 (special broadcast command) are not available for setup.

ESC ID setup steps are in Chapter 4.2.1.

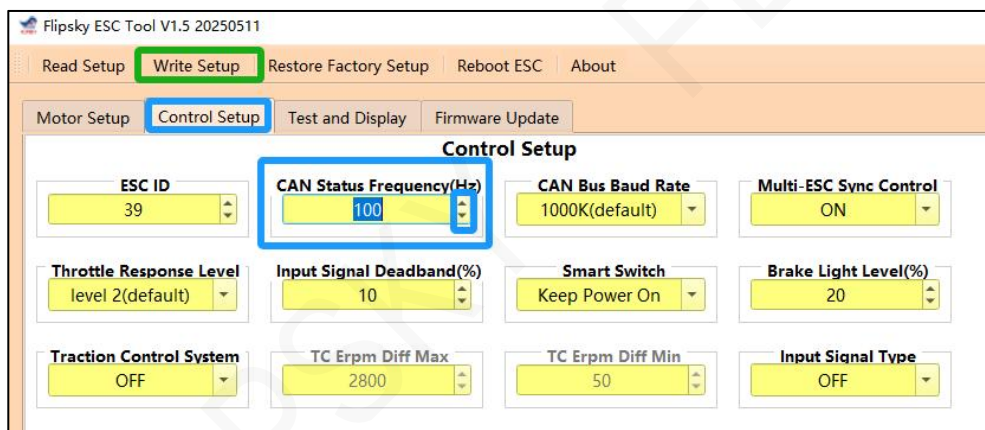
Fig.7-2.ESC ID



7.2 CAN Status Frequency Setup

CAN Status Frequency (Hz) : Each ESC will automatically return real-time data through the CAN bus, where the real-time data includes motor current, battery current, motor erpm, duty cycle, MOSFET temperature, motor temperature and battery voltage data. All of the above real-time data will be returned and sent to the CAN bus at regular intervals, and the frequency of sending can be set via CAN Status Frequency as Fig.7-3 below.

Fig.7-3.CAN status frequency

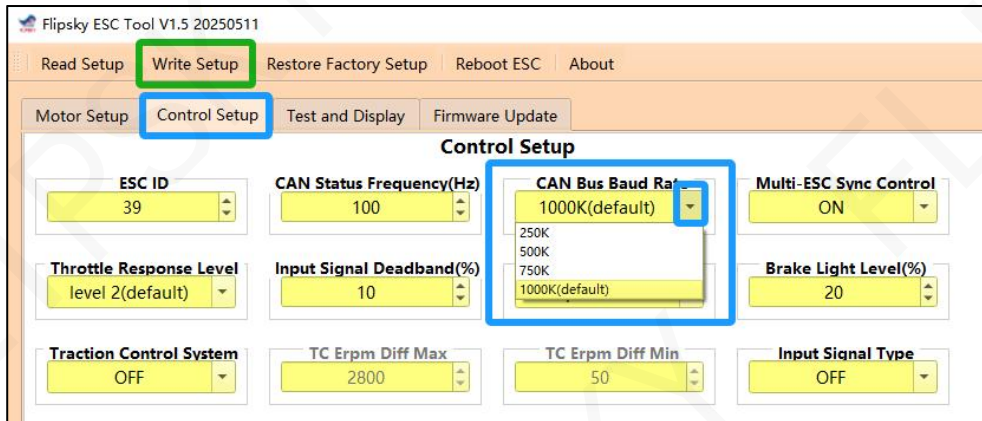


7.3 CAN Bus Baud Rate Setup

CAN Bus Baud Rate(Fig.7-4): Please make sure the consistency of CAN baud rate for the same CAN devices.

You can view the CAN bus baud rate setup in chapter 4.2.2.

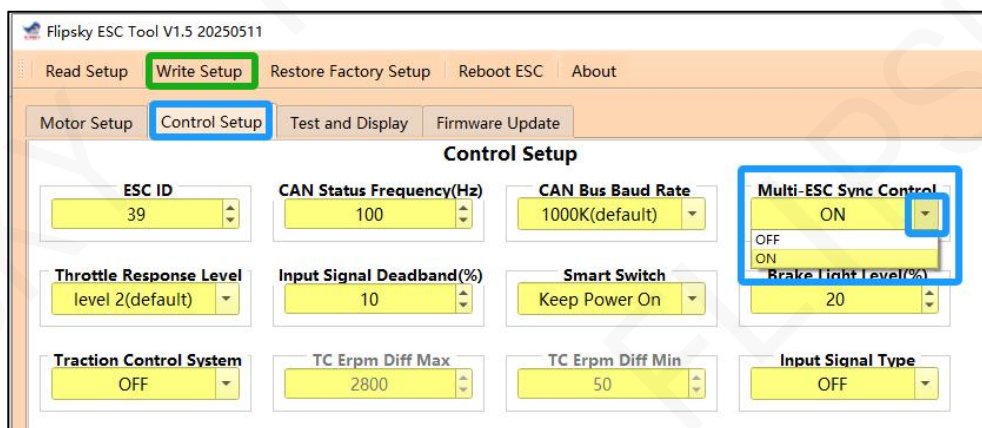
Fig.7-4.CAN bus baud rate



7.4 CAN Bus Multi-ESC Sync Control Setup

Multi-ESC Sync Control: synchronous control for multiple ESCs via CAN as Fig.7-5:

Fig.7-5.CAN Bus



Setup Steps:

- Click "Control Setup" to enter the control setup interface
- Click "Multi-ESC Sync Control" to select whether synchronous control of the CAN ESCs or

not. (ON - with synchronous control, OFF - not)

- Click the upper left menu "Write Setup" to save the setup

Parameter Specification:

When the ESC is used as master and "Multi-ESC Sync Control" is set to ON, the input signal will synchronize the master ESC and all slave ESCs mounted on the CAN bus; when "Multi-ESC Sync Control" is set to OFF, the input signal controls only the master ESC.

When the ESC acts as a slave and "Multi-ESC Sync Control" is set to ON, the slave ESC will accept control commands sent from the CAN bus; when "Multi-ESC Sync Control" is set to OFF, the slave ESC will block the control commands sent from the CAN bus.

Therefore, if you want multiple ESCs to be controlled synchronously, set all ESC's Multi-ESC Sync Control "ON".

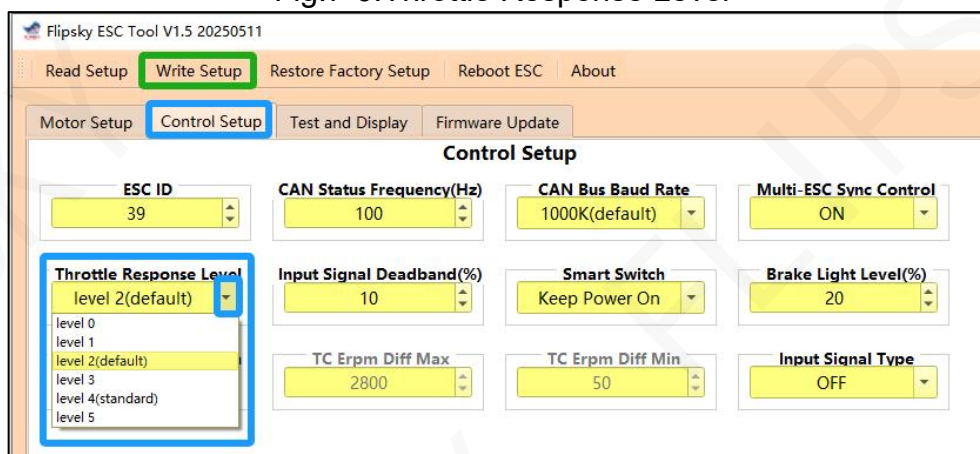
However, if you want to control each ESC separately in UART, PPM or ADC control type, set all ESC's Multi-ESC Sync Control "OFF".

If use your own control module as the master board and want to control the ESCs via CAN-Bus, set all ESC's Multi-ESC Sync Control "ON", and the Input Signal Type "OFF".

7.5 Throttle Response Level Setup

Throttle Response Level: Level 0~Level 5(as Fig.7-6):

Fig.7-6.Throttle Response Level



Setup Steps:

- Click "Control Setup" to enter the setup interface

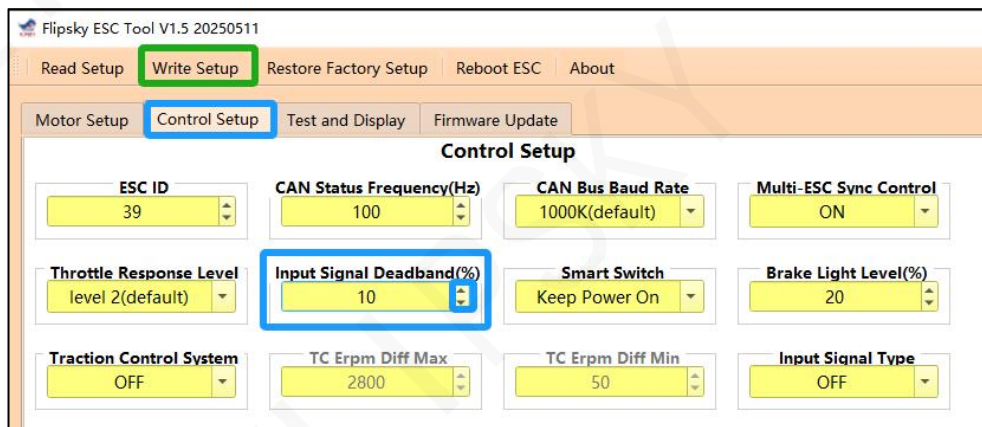
- Click "Throttle Response Level" to select the appropriate throttle response level
- Click "Write Setup" on the upper left to save the configuration

Parameter Specification: The higher of the level, the more sensitive of the throttle and brake response. Level 4 is the standard level which has no advance or lag on all input signals and is suitable for common users

7.6 Input Signal Deadband Setup

Input Signal Deadband: adjustable range: 1%~25%. as Fig.7-7:

Fig.7-7. Input signal deadband



Setup Steps:

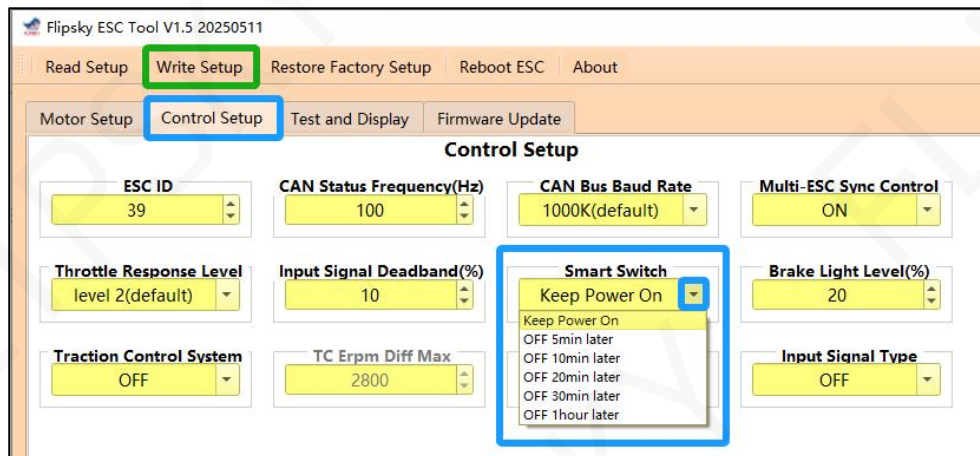
- Click "Control Setup" to enter the control setup interface
- Click "Input Signal Deadband" to adjust suitable input signal deadband
- Click "Write Setup" on the upper left to save the configuration

Parameter Specification: The ESC starts the current output only when the throttle input signal percentage is over than that of the input deadband setup . It's recommended to use the default value 10%. Regarding the input control type PPM and UART, the deadband percentage can be small or even 0% as the remote controller has already set it, for example, the Flipsky remote controller VX4 with 5% deadband already. It's strongly recommended to keep the default setting as 10% under ADC control.

7.7 Smart Switch

Smart Switch: This function only support FT85BD、FT60BD and FT85KS of FT series ESC. You can set it through Smart Switch as Fig.7-8:

Fig.7-8.Smart Switch setup

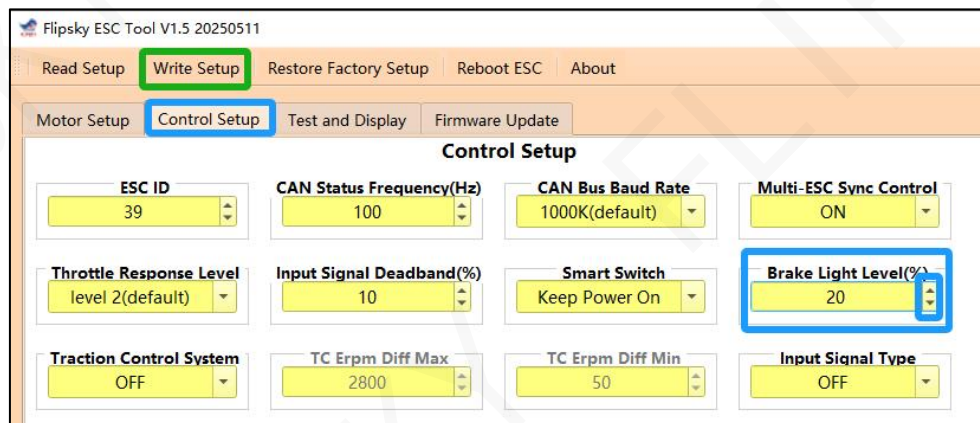


Parameter Specification: two modes: “Keep Power On” stands for always on without power-off. The second mode means power off after a specific time, for example, “OFF 5min later” supports auto shutoff without drive signal after 5 minutes later.

7.8 Brake Light Level Setup

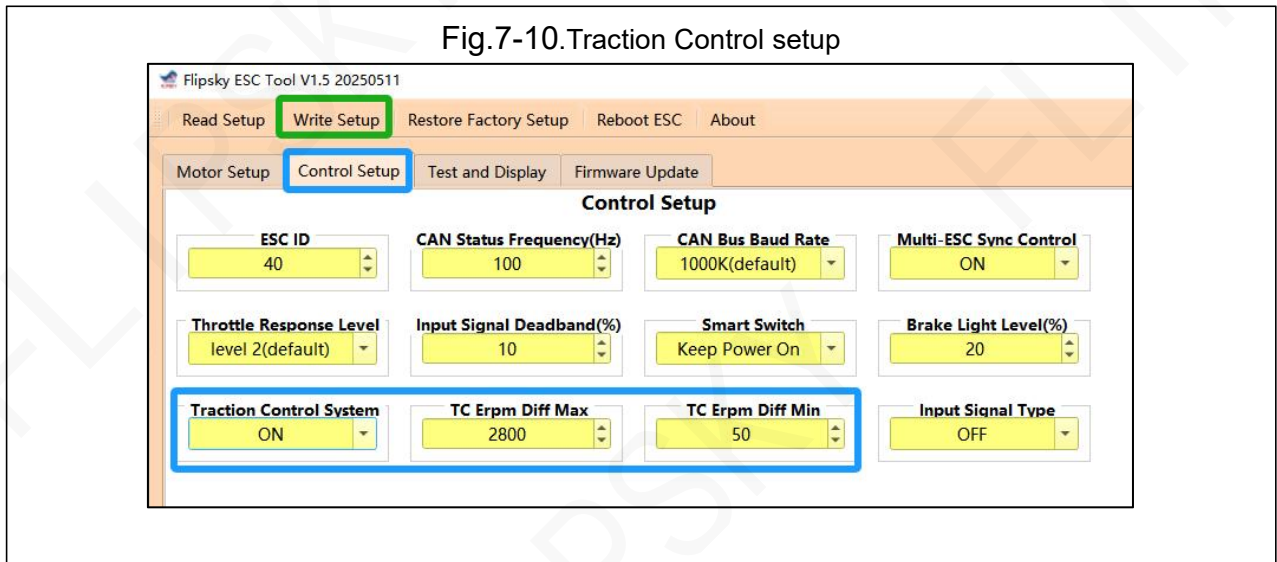
Brake Light Level: The brake light brightness setup in static. Adjustable range: 0~100%.

Fig.7-9.Brake light brightness setup in static



7.9 Traction Control Setup

TCR Control: Traction control is realized when multiple ESCs are jointly controlled, and the algorithm can be used to prevent the drive wheels from skidding when multiple ESCs accelerate the drive motors at the same time. This mode is only for current control (does not support current hybrid bi-directional control in PPM mode). Traction control setup as Fig.7-10:



■ Traction Control System

- ✧ ON
- ✧ OFF

Parameter Specification: This mode is mainly for preventing the drive wheel from slipping when ESK8 accelerates and maintains its stability.

■ TC Erpm Diff Max

Adjustable range: 0~10000

Parameter Specification: Maximum erpm difference threshold. Different motors correspond to different maximum erpm difference thresholds, which are typically set to 5% of the maximum erpm at full voltage. For example, a 6374 motor with a 7-pole KV of 160 would have a maximum erpm of about 56,500 at full voltage of 50.4V and 12S, so this value is set to $56,500 * 0.05 = 2825$.

■ TC Erpm Diff Min

Adjustable range: 0~1000

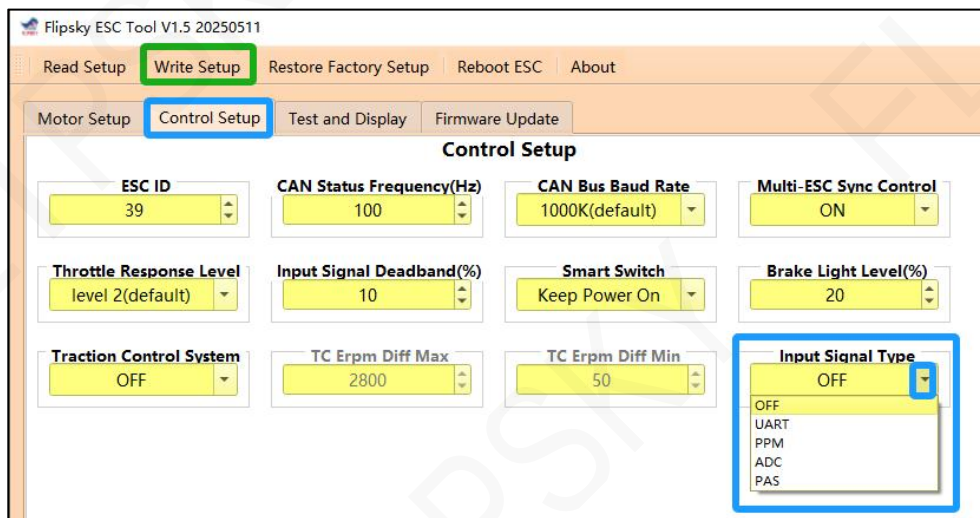
Parameter Specification: Minimum Erpm difference threshold. When the speed difference

is lower than this value, the TCR control will be cut off.

7.10 Input Signal Type Setup

Input Signal Type:OFF/UART/PPM/ADC/PAS. Default as OFF(as Fig.7-11). View the chapter below for details.

Fig.7-11.Input signal type setup



7.10.1 OFF

When “Input Signal Type” is set to OFF(Fig.7-12), all input signals will be shielded. You can only control the ESC through Current, Duty, Brake and other buttons in Test and Display interface(Fig.7-13). View button operations in Chapter 8.1.1.

Fig. 7-12.OFF

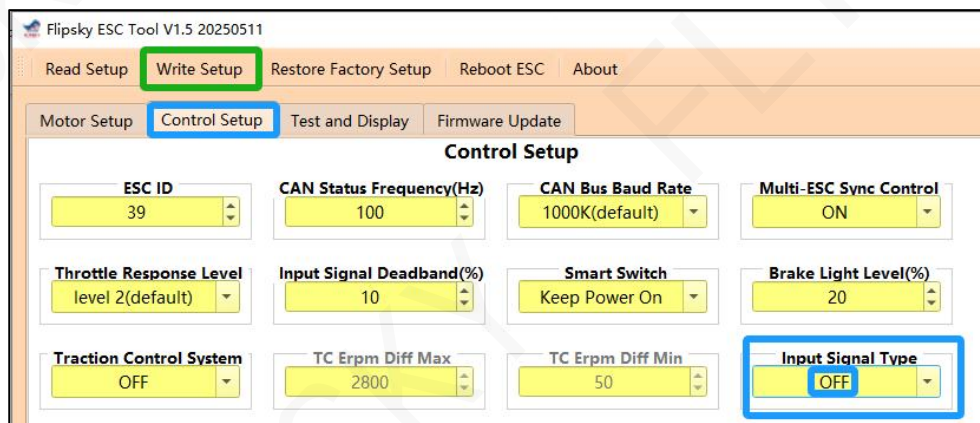
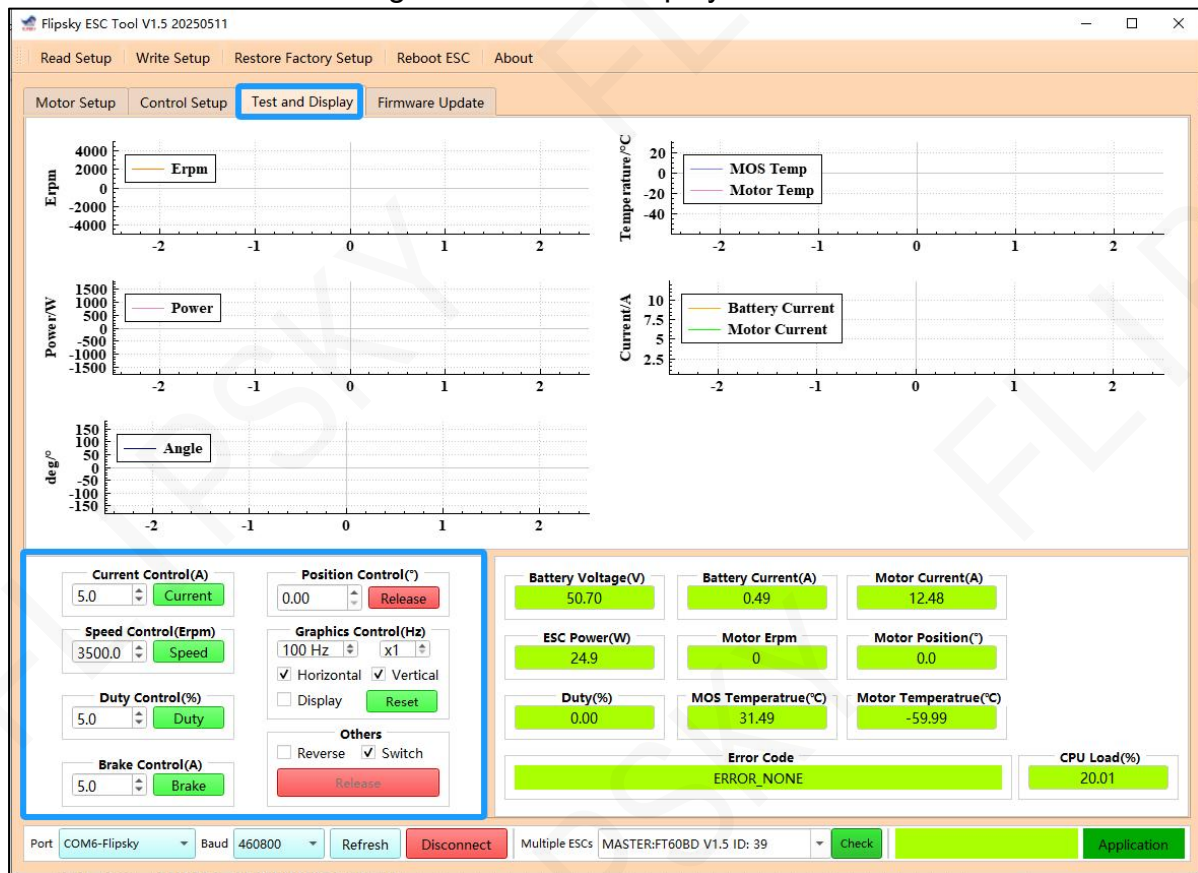


Fig.7-13.Test and Display interface



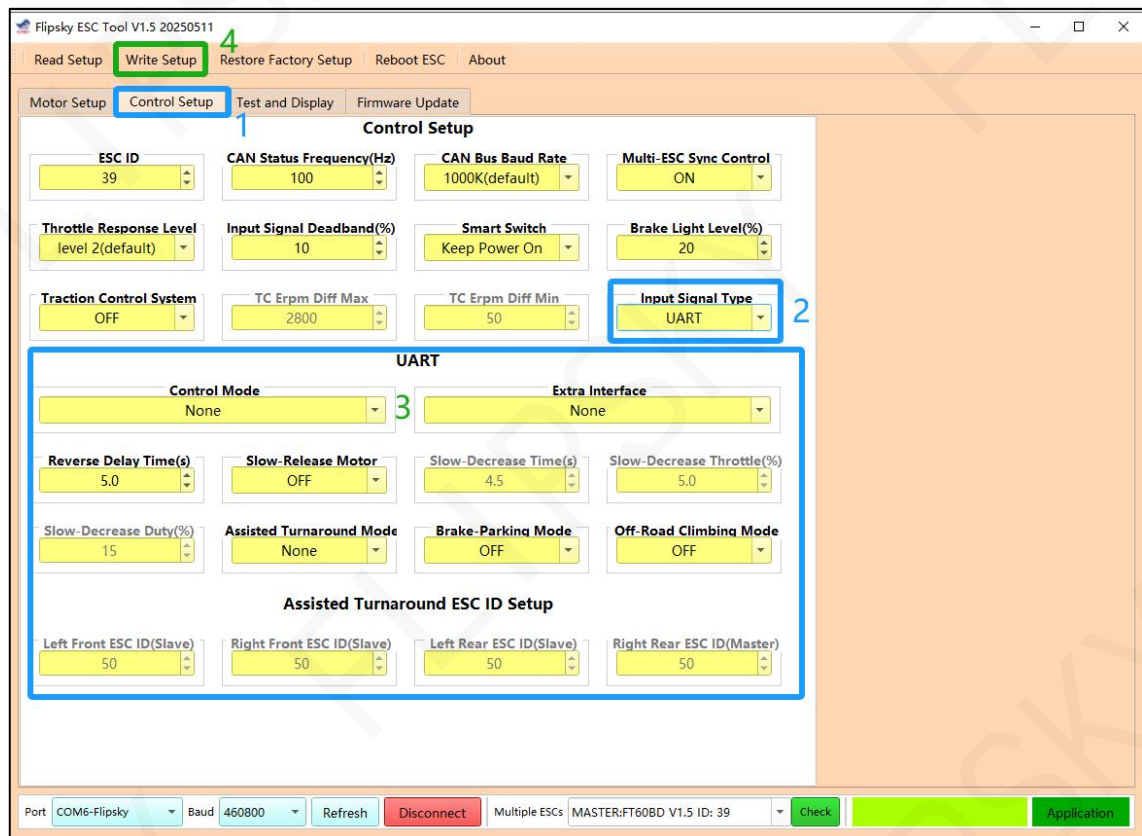
7.10.2 UART

7.10.2.1 UART Parameter Specification and Setup

UART control mode can support all FT series remotes.(VX1, VX1Pro, VX2, VX3, VX3Pro, VX4, VX4Pro, VX5) Support limit its maximum rpm in different gears.

Setup steps are as Fig.7-14 below when use UART signal to control the ESC:

图 7-14.UART 信号控制设置步骤



NOTE: Step 3 and Step 10 is must, other parameters can be set according to your needs.

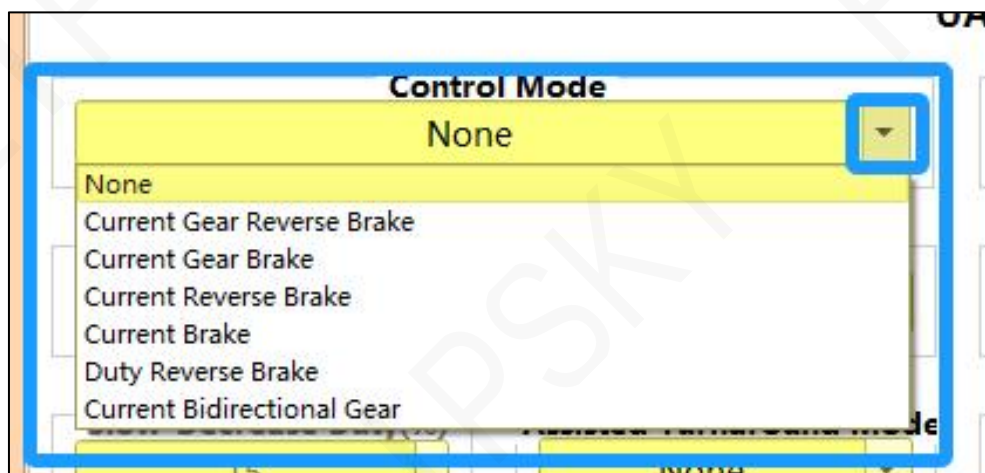
- Step 1: Click "Control Setup" to enter the control setup interface
- Step 2: Select UART in "Input Signal Type"
- Step 3: Select suitable UART Control Mode accordingly(view details in 7.10.2.2)
- Step 4: Set UART Reverse Delay Time if you need (view details in 7.10.2.3)
- Step 5: Set Extra Interface if you need(view details in 7.10.2.4)
- Step 6: Set Slow-Release Motor if you need (view details in 7.10.2.5)
- Step 7: Set Assisted Turnaround Mode and Assisted Turnaround ESC ID Setup if you need(view details in 7.10.2.6)

- Step 8: Set Brake Parking Mode if you need(view details in 7.10.2.7)
- Step 9: Set Off-Road Climbing Mode if you need(view details in 7.10.2.8)
- Step 10: Click “Write Setup” on the upper left again, write and save the configuration to ESC.

7.10.2.2 UART Control Mode Setup

User can select applicable UART Control Mode as Fig.7-15:

图 7-15.UART Control Mode



- None
- Current Gear Reverse Brake: Current control mode: gear speed limit, reverse with brake, reverse
- Current Gear Brake: Current control mode: gear speed limit, reverse with brake
- Current Reverse Brake: Current control mode: reverse with brake, reverse
- Current Brake: Current control mode: reverse with brake
- Duty Gear Brake Reverse: Duty cycle control mode: gear speed level limit, reverse with brake, reverse
- Current Bidirectional Gear: Current control mode: bilateral control, gear speed limit

7.10.2.3 UART Reverse Delay Time Setup

The UART control mode supports reverse function, so you can set the UART Reverse Delay Time per your needs. The adjustable range: 0.0s~10s, and the default is 5s as shown in 7-16:

Fig.7-16.UART Reverse Delay Time Setup

UART			
Control Mode None	Extra Interface None		
Reverse Delay Time(s) 5.0	Slow-Release Motor OFF	Slow-Decrease Time(s) 4.5	Slow-Decrease Throttle(%) 5.0
Slow-Decrease Duty(%) 15	Assisted Turnaround Mode None	Brake-Parking Mode OFF	Off-Road Climbing Mode OFF

Parameter Specification: When the motor is stopped, pull the throttle back to the maximum and then start to rotate in the reverse direction. It can reach the maximum reverse speed within the set time period(as Fig.7-16, it's 5s as set)

7.10.2.4 Extra Interface Setup

This function only support FT85BD, FT60BD and FT85KS of FT series ESC. Support front light, brake light and horn control under UART control. Support VX2, VX3, VX4, VX4Pro and VX5 remote to control front light, brake light and horn. User can set through Extra Interface accordingly as Fig.7-17:

Fig.7-17.Extra Interface

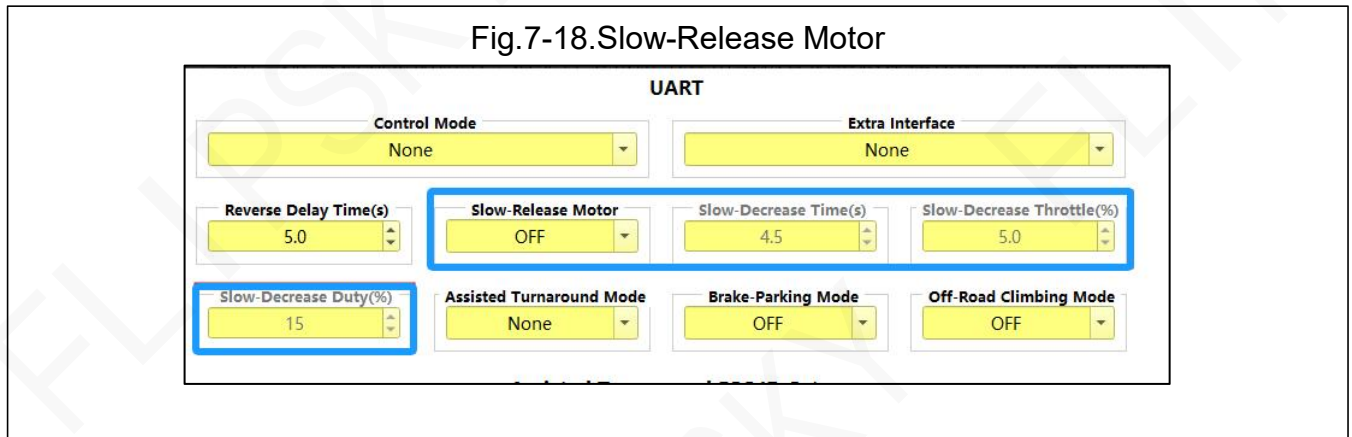
UART	
Control Mode None	Extra Interface None
Reverse Delay Time(s) 5.0	Slow-Release Motor OFF
Slow-Decrease Duty(%) 15	Assisted Turnaround Mode None

- None
- Front Light
- Brake Light
- Horn
- Front and Brake Light
- Front and Horn
- Brake Light and Horn
- Front, Brake Light and Horn

7.10.2.5 Slow-Release Motor Setup

Slow-Release Motor function enables FT series ESCs to maintain a certain current output after releasing the throttle under the UART input signal, which can offset the resistances of high gear ratios (>3.5:1) off-road skateboards or other sports kits and put the skateboards or sports kits in a release state with almost zero resistance and smoother starting process.

The slow-release setup steps are as Fig.7-18 below:



User can set below parameters accordingly.

Parameter Specification:

- **Slow-Release Motor:** Setup column, default: OFF
- **Slow-Decrease Time:** Current output releasing time, default: 4.5s
- **Slow-Decrease Throttle:** Current output throttle percentage, default:5% ($0.05 \times \text{Motor Current}$)
- **Slow-Decrease Duty:** When the motor duty cycle is lower than this value, this function is off. Only when the duty cycle is higher than this value will this function be effective.

NOTE: This function is not available under the control mode of Duty Gear Brake Reverse (Duty cycle control mode: gear speed limit, reverse with brake, reverse)

7.10.2.6 Assisted Turnaround Mode and ESC ID Setup

Assisted Turnaround: This assisted turnaround function is applicable in UART control only. Also, it only supports VX4, VX4Pro and VX5 remote. (VX4 shall be with the latest firmware VX4_V1_4 or above) View detailed operations in user manuals of VX4, VX4Pro and VX5.

Setup Steps:

- Step 1: Select Assisted Turnaround Mode accordingly as Fig.7-19:

Fig.7-19.Assisted Turnaround Mode

The screenshot shows the 'UART' configuration window. The 'Assisted Turnaround Mode' dropdown is highlighted with a blue box, showing options: None, Dual Drive Mode, and Quad Drive Mode. Other settings include Control Mode (None), Extra Interface (None), Reverse Delay Time (5.0s), Slow-Release Motor (OFF), Slow-Decrease Time (4.5s), Slow-Decrease Throttle (5.0%), Slow-Decrease Duty (15%), Brake-Parking Mode (OFF), Off-Road Climbing Mode (OFF), and four ESC ID fields (Left Front, Right Front, Left Rear, Right Rear) all set to 50.

Parameter Specification:

- Assisted Turnaround Mode:
 - ✧ None: Turn off assisted turnaround mode.
 - ✧ Dual Drive Mode: Dual drive turnaround mode. Uses only two ESC drives. One forward and one reverse.
 - ✧ Quad Drive Mode: 4WD turnaround mode. All four ESCs drive at the same time. Forward on the same side, reverse on the other side.

➤ Step 2: Set assisted turnaround ESC ID as Fig.7-20:

Fig.7-20. Assisted turnaround ESC ID Setup

The screenshot shows the 'UART' configuration window. The 'Assisted Turnaround Mode' is set to 'Quad Drive Mode'. The 'Assisted Turnaround ESC ID Setup' section is highlighted with a blue box, showing four ESC ID fields: Left Front ESC ID(Slave), Right Front ESC ID(Slave), Left Rear ESC ID(Slave), and Right Rear ESC ID(Master), all set to 50. Other settings are the same as in Fig.7-19.

- Assisted Turnaround ESC ID Setup:
 - Left Front ESC ID(Slave): Set in 4WD.
 - Right Front ESC ID(Slave): Set in 4WD.

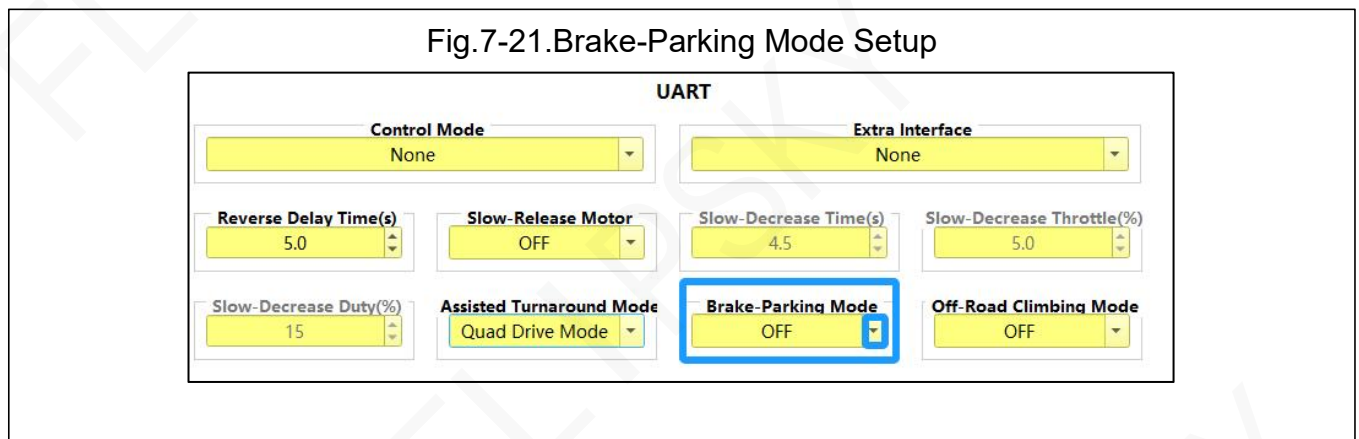
- Left Rear ESC ID(Slave): Set in 4WD.
- Right Rear ESC ID(Master): Set in 4WD.

NOTE:

- (1) Dual Drive Mode: Set “Left Rear ESC ID(Slave)” and “Right Rear ESC ID(Master)”.
- (2) Quad Drive Mode: Set “Left Front ESC ID(Slave)”, “Right Front ESC ID(Slave)”, “Left Rear ESC ID(Slave)” and “Right Rear ESC ID(Master)”.

7.10.2.7 Brake-Parking Mode Setup

Brake-Parking Mode: Pull the remote controller throttle gear/bar 0.5S to enter into Brake-Parking Mode after the motor stop running. The greater percentage of pulling back the throttle gear, the bigger of the parking force. Setup as shown in Fig.7-21:



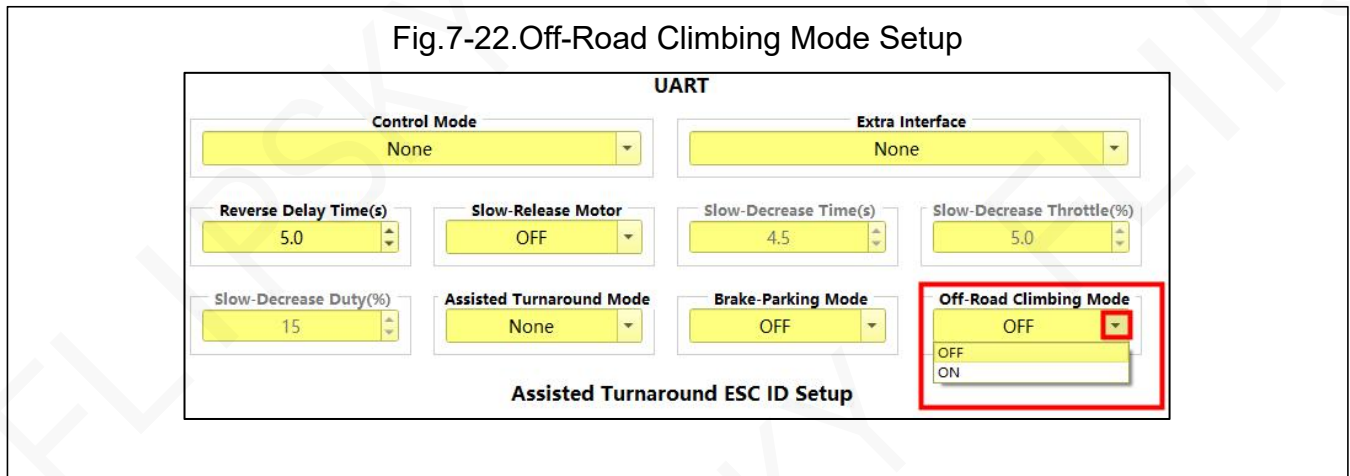
NOTE: This mode injects current into the motor d-axis to make the coil generate magnetic force and attract the rotor to achieve the parking effect, so current will be consumed when parking. The maximum d-axis current can reach Motor Current Max when parking. It is not recommended to park in this mode for a long time, otherwise the motor will get hot and lead to overload current limiting phenomenon.

When in Reverse Mode like Current Gear Reverse Brake or Current Gear Brake mode, max pulling the throttle gear / bar will end the Brake-Parking Mode but enter into Reverse Mode instead.

Currently, this mode supports for remote VX4(firmware version: V1.4. or above), VX4Pro(firmware version: V1.0 or above), VX2(firmware version:V1.9), VX3(firmware version:V1.9) only. Please check user manuals of VX2, VX3, VX4 and VX4Pro for details.

7.10.2.8 Off-Road Climbing Mode

Off-Road Climbing Mode: In this mode, all CAN ESCs will be switched to duty cycle control and allow users to pass off-road passing gravel roads or climbing high slopes easily by controlling speed of the four wheels synchronously via remote controller. Setup as shown in Fig.7-22:



NOTE:

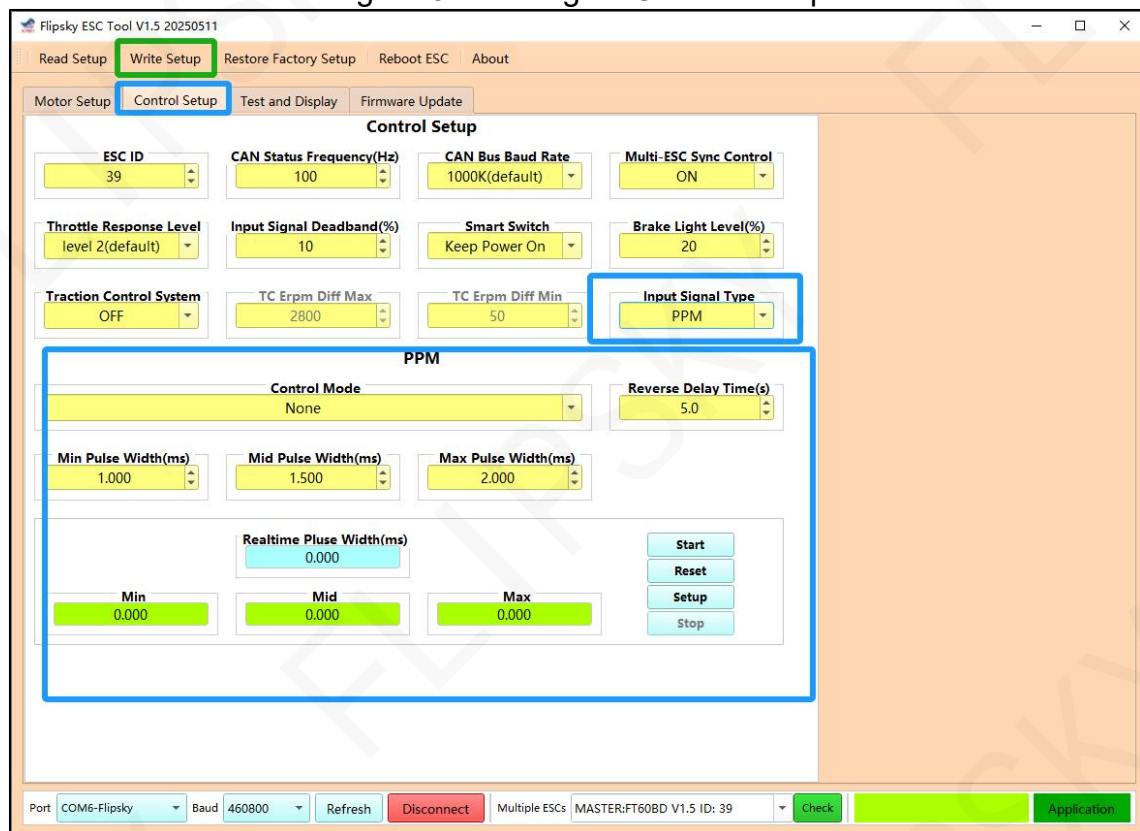
1. After using this mode, the reverse mode will be disabled.
2. At present, this mode supports for remote VX4(firmware version: V1.4.), VX4Pro(firmware version: V1.0), VX2(firmware version:V1.9), VX3(firmware version:V1.9) only. Please check user manuals of VX2, VX3, VX4 and VX4Pro for details.

7.10.3 PPM

7.10.3.1 PPM Parameter Specification and Setup

The PPM signal pulse width usually ranges from 1ms to 2ms, so theoretically the PPM control frequency supports up to 500Hz. Considering the MCU clock accuracy, it is recommended to use the PPM control frequency less than 400Hz. Setting steps as Fig.7-23 below:

Fig.7-23.PPM Signal Control Setup



- Step 1: Click "Control Setup" to enter the setup interface
- Step 2: Switch the "Input Signal Type" to "PPM".
- Step 3: Calibrate when use PPM as input signal for the first time(check details in chapter 7.10.3.2)

NOTE: Calibrate the PPM signal for one side of the dual ESC controller is workable. The other side's Input Signal Type set "OFF".

- Step 4: Select the appropriate PPM Control Mode as you need (see details in 7.10.3.3)
- Step 5: Set the Reverse Delay Time (see details in 7.10.3.4)
- Step 6: Click the upper left menu "Write Setup" to save the setup.

7.10.3.2 PPM Signal Calibration Steps

Calibrate PPM signal first when use it as input signal for the first time. Please plug the PPM signal cable to the ESC PPM port before calibration. Take VX1 remote of Flipsky as an example:

- Click "Start" to begin the calibration as Fig.7-24 below:

Fig.7-24.Enter PPM signal calibration

The screenshot shows the PPM signal calibration interface. At the top, there are three input fields for pulse widths: "Min Pulse Width(ms)" with a value of 1.000, "Mid Pulse Width(ms)" with a value of 1.500, and "Max Pulse Width(ms)" with a value of 2.000. Below these, there is a "Realtime Pluse Width(ms)" field showing 1.497. At the bottom, there are three green boxes for "Min" (1.496), "Mid" (1.497), and "Max" (1.499). On the right side, there are four buttons: "Start", "Reset", "Setup", and "Stop". The "Start" button is highlighted with a blue border.

NOTE: No PPM signal can be detected if the remote receiver hasn't been plugged with the ESC PPM wire when you click "Start" as Fig7-25:

Fig.7-25.No PPM signal

The screenshot shows the PPM signal calibration interface when no signal is detected. The "Realtime Pluse Width(ms)" field shows 0.000. The "Min", "Mid", and "Max" green boxes at the bottom all show 0.000. The "Start", "Reset", "Setup", and "Stop" buttons are on the right side.

- Connect the remote receiver with the ESC PPM wire in this case. If the PPM signal data seems abnormal (as shown in Fig.7-26), please click "Reset". (Refer to Fig.7-24).

Fig.7-26.Abnormal PPM signal

The screenshot shows the PPM signal calibration interface with abnormal signal data. The "Realtime Pluse Width(ms)" field shows 1.497. The "Min", "Mid", and "Max" green boxes at the bottom show 0.000, 0.749, and 1.498 respectively. The "Reset" button on the right side is highlighted with a blue border.

- Pull and push the remote throttle gear to refresh the minimum, middle and maximum PPM signal values as Fig.7-27, Fig.7-28:

Fig.7-27.Refresh the maximum PPM signal

Min Pulse Width(ms)	Mid Pulse Width(ms)	Max Pulse Width(ms)
1.000	1.500	2.000

Realtime Pluse Width(ms)
1.992

Min	Mid	Max
1.496	1.746	1.995

Start

Reset

Setup

Stop

Fig.7-28.Refresh the minimum PPM signal

Min Pulse Width(ms)	Mid Pulse Width(ms)	Max Pulse Width(ms)
1.000	1.500	2.000

Realtime Pluse Width(ms)
1.000

Min	Mid	Max
0.999	1.497	1.995

Start

Reset

Setup

Stop

- Click “Setup” to write the current values into Min Pulse Width, Mid Pulse Width and Max Pulse Width separately as Fig.7-29:

Fig.7-29.Write current PPM signal value

Min Pulse Width(ms)	Mid Pulse Width(ms)	Max Pulse Width(ms)
0.999	1.497	1.995

Realtime Pluse Width(ms)
1.497

Min	Mid	Max
0.999	1.497	1.995

Start

Reset

Setup

Stop

- Click "Stop" to complete the PPM calibration as Fig.7-30:

Fig.7-30.PPM signal calibration ends

The screenshot shows the PPM calibration interface. At the top, it says 'PPM'. Below this, there are several fields: 'Control Mode' set to 'Current Reverse(reverse:1.0~1.5ms,forward:1.5~2.0ms)', 'Reverse Delay Time(s)' set to 0.5, 'Min Pulse Width(ms)' set to 1.000, 'Mid Pulse Width(ms)' set to 1.499, and 'Max Pulse Width(ms)' set to 1.994. Below these, there is a 'Realtime Pluse Width(ms)' field set to 1.499. At the bottom, there are three green boxes labeled 'Min' (1.000), 'Mid' (1.499), and 'Max' (1.994). On the right side, there are four buttons: 'Start', 'Reset', 'Setup', and 'Stop'. The 'Stop' button is highlighted with a blue border.

7.10.3.3 PPM Control Mode Setup

User can select suitable PPM Control Mode accordingly as Fig.7-31:

Fig.7-31.PPM Control Mode

The screenshot shows the PPM Control Mode setup interface. At the top, it says 'PPM'. Below this, there is a 'Control Mode' dropdown menu that is open, showing a list of options: 'None', 'Current(forward:1.0~2.0ms)', 'Current Brake(brake:1.0~1.5ms,forward:1.5~2.0ms)', 'Current Reverse(reverse:1.0~1.5ms,forward:1.5~2.0ms)', 'Current Reverse Brake(reverse/brake:1.0~1.5ms,forward:1.5~2.0ms)', 'Duty(forward:1.0~2.0ms)', 'Duty Reverse(reverse:1.0~1.5ms,forward:1.5~2.0ms)', 'Duty Reverse Brake(reverse/brake:1.0~1.5ms,forward:1.5~2.0ms)', 'Duty Reverse Slow(reverse:1.0~1.5ms,forward:1.5~2.0ms)', and 'Speed(forward:1.0~2.0ms)'. Below the dropdown menu, there are three green boxes labeled 'Min' (0.000), 'Mid' (0.000), and 'Max' (0.000). On the right side, there are four buttons: 'Start', 'Reset', 'Setup', and 'Stop'. The 'Control Mode' dropdown menu is highlighted with a blue border.

- None
- Current
 - Current control mode, unidirectional control, forward only:1.0~2.0ms,
- Current Brake
 - Current control mode, bidirectional control, brake:1.0~1.5ms,forward:1.5~2.0ms
- Current Reverse
 - Current control mode, bidirectional control, no brake, reverse:1.0~1.5ms, forward:1.5~2.0ms
- Current Reverse Brake
 - Current control mode, bidirectional control, brake/reverse:1.0~1.5ms, forward:1.5~2.0ms.

- Duty(forward:1.0~2.0ms)

Current control mode, bidirectional control, brake/reverse:1.0~1.5ms, forward:1.0~2.0ms.

- Duty Reverse

Duty cycle control mode, bidirectional control, backward:1.0~1.5ms, forward:1.5~2.0ms.

- Duty Reverse Brake

Duty cycle control mode, bidirectional control, brake/reverse:1.0~1.5ms, forward:1.5~2.0ms.

- Duty Reverse Slow

Duty cycle control mode, bidirectional control, reverse:1.0~1.5ms, forward:1.5~2.0ms.

NOTE: No back EMF is generated when the accelerator is released.

- Speed

PID control mode, unidirectional control, forward only:1.0~2.0ms

- Speed Reverse

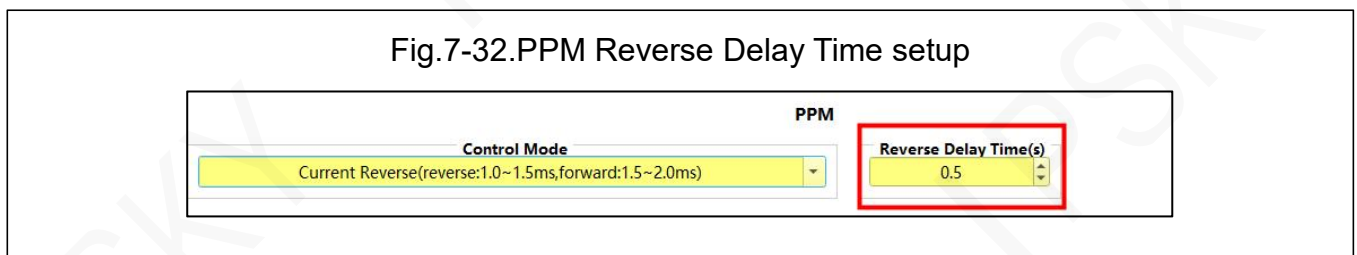
PID control mode, bidirectional control, reverse:1.0~1.5ms, forward:1.5~2.0ms

- Current Hybrid Bidirectional With Brake

Bidirectional current mixed control, reverse/brake: 1.0~1.5ms, forward/brake: 1.5~2.0ms

7.10.3.4 PPM Reverse Delay Time Setup

The PPM control mode supports reverse function, so you can set the UART Reverse Delay Time as you need. Adjustable range: 0.0s~10s, and the default is 5s as Fig.7-32:

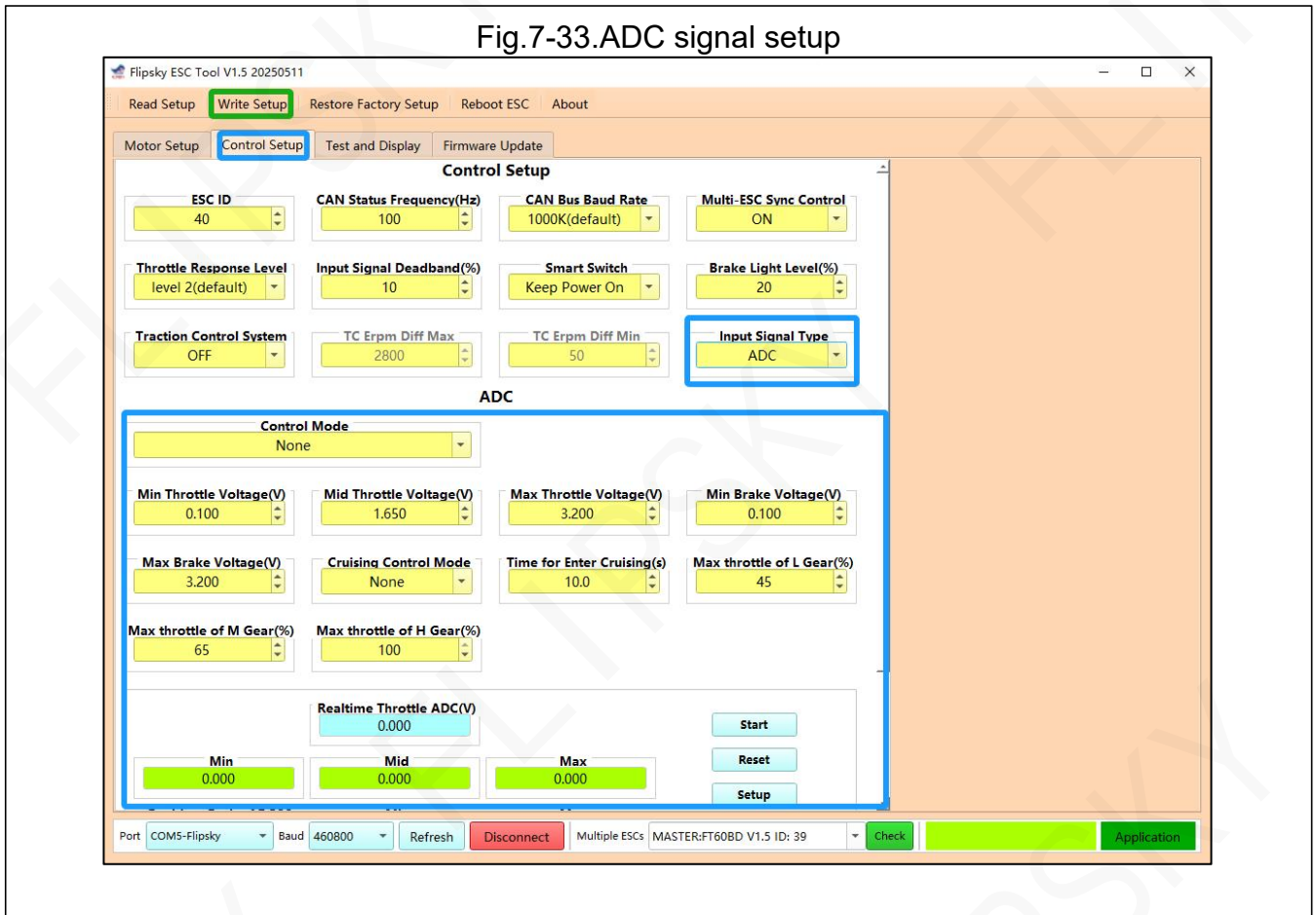


Parameter Specification: When the motor is stopped, pull the throttle back to the maximum and start to rotate in the reverse direction. The reverse speed can reach the maximum within the set time period (as shown in Fig.7-32, it's 5 seconds as this set).

7.10.4 ADC

7.10.4.1 ADC Parameter Specification and Setup

ADC control type voltage range: 0V~3.3V. Please make sure the ADC input voltage lower than 3.3V to avoid the ESC 3.3V chip damage. Setting steps are as Fig.7-33. below:



- Step 1: Click "Control Setup" to enter the setup interface
- Step 2: Switch the "Input Signal Type" to "ADC"
- Step 3: Do ADC throttle calibration for the first time as details in Chapter 7.10.4.2

NOTE: When using a dual-drive ESC, only select the signal control type calibration for the signal master or slave under control, and set the signal type of the other master/ slave to OFF.

- Step 4: Select the appropriate ADC Control Mode per your needs (see details in chapter 7.10.4.3)
- Step 5: Select the Cruising Control Mode per your needs (view details in chapter 7.10.4.4)
- Step 6: Set the Time for Enter Cruising (view details in chapter 7.10.4.4)
- Step 7: Set the maximum throttle percentage per your needs(view details in chapter 7.10.4.4)

7.10.4.5)

- Step 8: Click the upper left menu "Write Setup" to save the setup.

7.10.4.2 ADC Input Throttle Calibration

Do ADC throttle calibration when use it as input signal for the first time as follows:

- Click "Start" to begin the calibration as Fig.7-34 below:

Fig.7-34.Enter ADC input throttle calibration

Realtime Throttle ADC(V) 0.826			Start Reset Setup Stop
Min 0.825	Mid 0.826	Max 0.828	
Realtime Brake ADC(V) 0.853	Min 0.852	Max 0.854	

- Twist the ADC throttle to refresh the min, mid and max throttle signal as Fig.7-35 below:

Fig.7-35.Refresh ADC throttle signal

Realtime Throttle ADC(V) 0.824			Start Reset Setup Stop
Min 0.823	Mid 1.670	Max 2.518	
Realtime Brake ADC(V) 0.853	Min 0.849	Max 0.853	

- Press the ADC brake to refresh the min, mid and max brake value as Fig.7-36 below:

Fig.7-36. Refresh ADC brake signal

Realtime Throttle ADC(V) 0.824			Start Reset Setup Stop
Min 0.820	Mid 1.669	Max 2.518	
Realtime Brake ADC(V) 0.849	Min 0.847	Max 2.559	

- Click "Setup" to write the min and max ADC signal value to Min Throttle Voltage, Mid Throttle Voltage, Max Throttle Voltage, Min Brake Voltage and Max Brake Voltage separately as Fig.7-37 below:

Fig.7-37. Write current ADC configurations

Min Throttle Voltage(V) 0.826	Mid Throttle Voltage(V) 1.669	Max Throttle Voltage(V) 2.513	Min Brake Voltage(V) 0.842
Max Brake Voltage(V) 2.549	Cruising Control Mode None	Time for Enter Cruising(s) 10.0	Max throttle of L Gear(%) 45
Max throttle of M Gear(%) 65	Max throttle of H Gear(%) 100		
Realtime Throttle ADC(V) 0.828			
Min 0.826	Mid 1.669	Max 2.513	
Realtime Brake ADC(V) 0.852	Min 0.842	Max 2.549	

Start
Reset
Setup
Stop

- Click "Stop" to finish the calibration as Fig.7-38 as below:

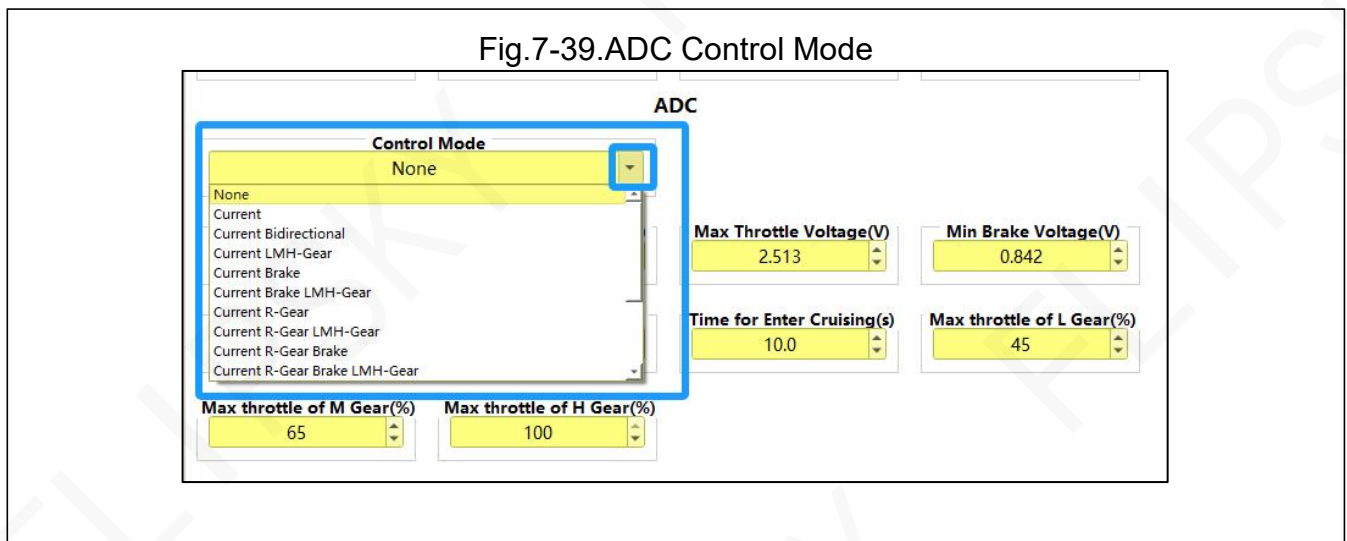
Fig.7-38. ADC input setup complete

Realtime Throttle ADC(V) 0.822			
Min 0.820	Mid 1.669	Max 2.518	
Realtime Brake ADC(V) 0.846	Min 0.845	Max 2.559	

Start
Reset
Setup
Stop

7.10.4.3 ADC Control Mode Setup

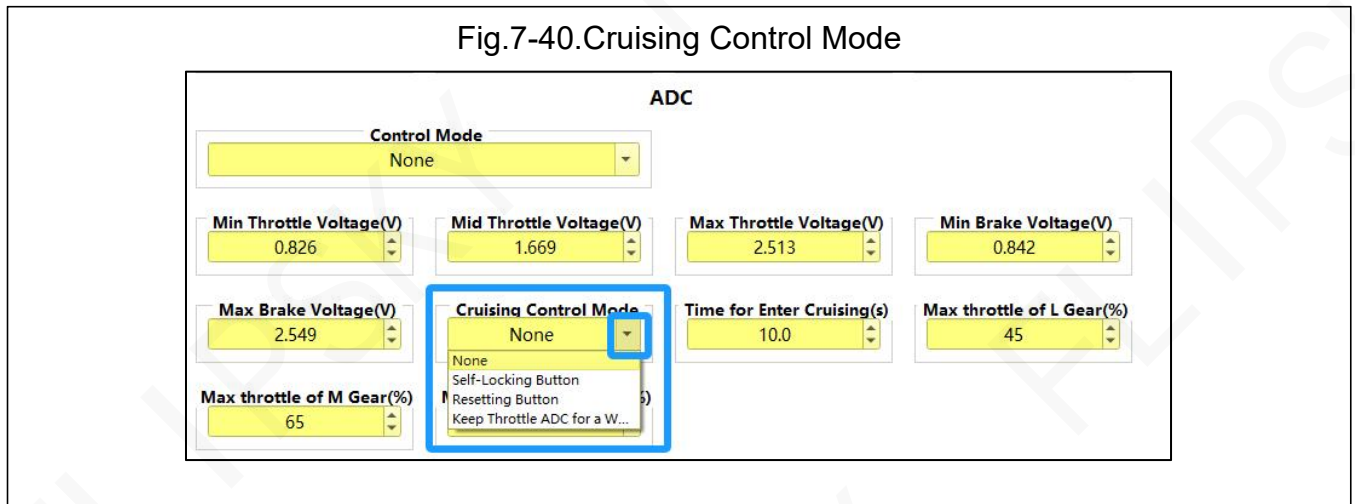
Select the appropriate ADC Control Mode per your needs as Fig.7-39 below:



- None: No response to any ADC input signal.
- Current: Current control mode, ADC throttle only without brake.
- Current Bidirectional: Current control mode, enable Throttle ADC and bidirectional current control
- Current LMH-Gear: Current control mode, only enable ADC throttle and L\M\H speed gear.
- Current Brake: Current control mode, enable ADC throttle and brake.
- Current Brake LMH-Gear: Current control mode, enable Throttle ADC, Brake ADC and L\M\H speed gear
- Current R-Gear: Current control mode, enable Throttle ADC and reverse gear.
- Current R-Gear LMH-Gear: Current control mode, enable Throttle ADC, reverse gear and L\M\H speed gear
- Current R-Gear Braker: Current control mode, enable Throttle ADC, reverse gear and Brake ADC
- Current R-Gear Brake LMH-Gear: Current control mode, enable Throttle ADC, reverse gear , Brake ADC and L\M\H speed gear
- Duty: Duty cycle mode, ADC throttle only without brake
- Duty Bidirectional: Bidirectional duty cycle control mode, ADC throttle only without brake
- Duty R-Gear: Reverse duty cycle control mode, Throttle ADC only without brake
- Speed: PID control mode, Throttle ADC only without brake
- Speed Bidirectional: Bidirectional PID control mode, Throttle ADC only without brake
- Speed R-Gear: Reverse PID control mode, Throttle ADC only without brake

7.10.4.4 Cruising Control Mode Setup

Cruising Control Mode: three modes available, select different mode accordingly as Fig.7-40:



Parameter Specification:

■ None

Turn off Cruising Control Mode

■ Self-Locking Button

Using self-locking button to activate the cruising control. Connect the PPM and GND pins for the self-locking button. Press the switch button to enter into cruise mode, and then re-press the button to bump out to end the cruise mode.

NOTE:

- The cruise mode can be entered only in status of throttle signal detected or motor running with rpm
- The cruise mode will stop if receive a brake signal in ADC control mode

■ Resetting Button

Using resetting button to activate the cruising control. Connect the PPM and GND pins for the self-locking button. Press once to enter into the cruise mode, and press again to end the cruise mode.

NOTE:

- The cruise mode can be entered only in status of throttle signal detected or motor running with rpm
- The cruise mode will stop if receive a brake signal in ADC control mode.

■ Keep Throttle ADC for a While

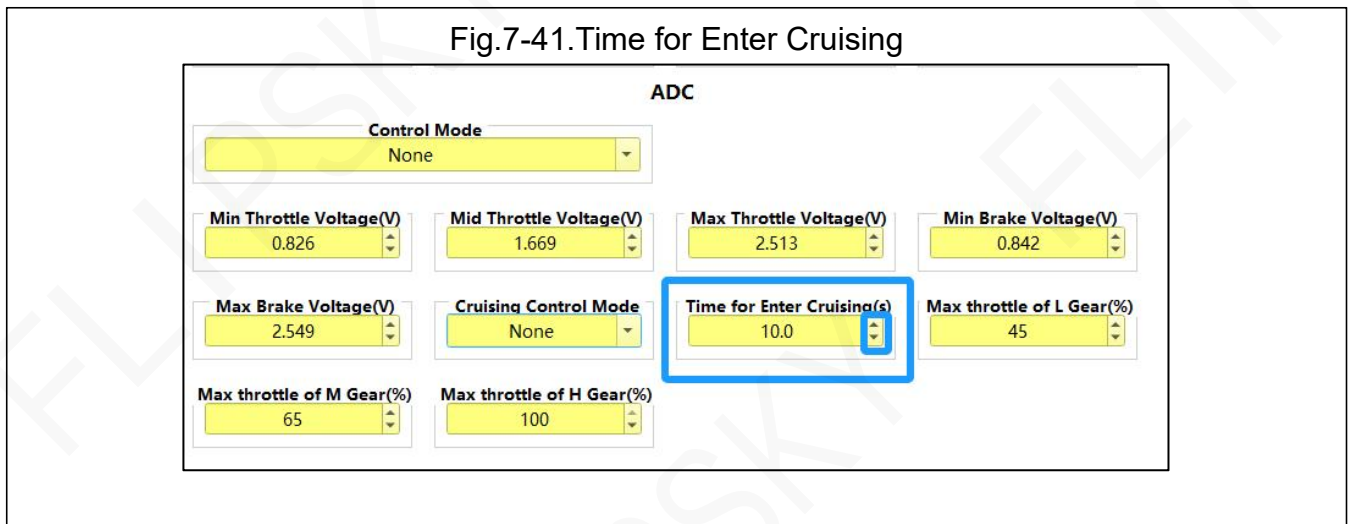
Entering the cruising control by keeping Throttle ADC for a while. Set the time via **Time for Enter Cruising**. Keep the throttle at desired level for a period of time and release, and then you will enter the cruise mode and stay. If any throttle change or brake signal is detected, the mode will be exited.

NOTE:

- The cruise mode will stop if receive a brake signal in ADC control mode.

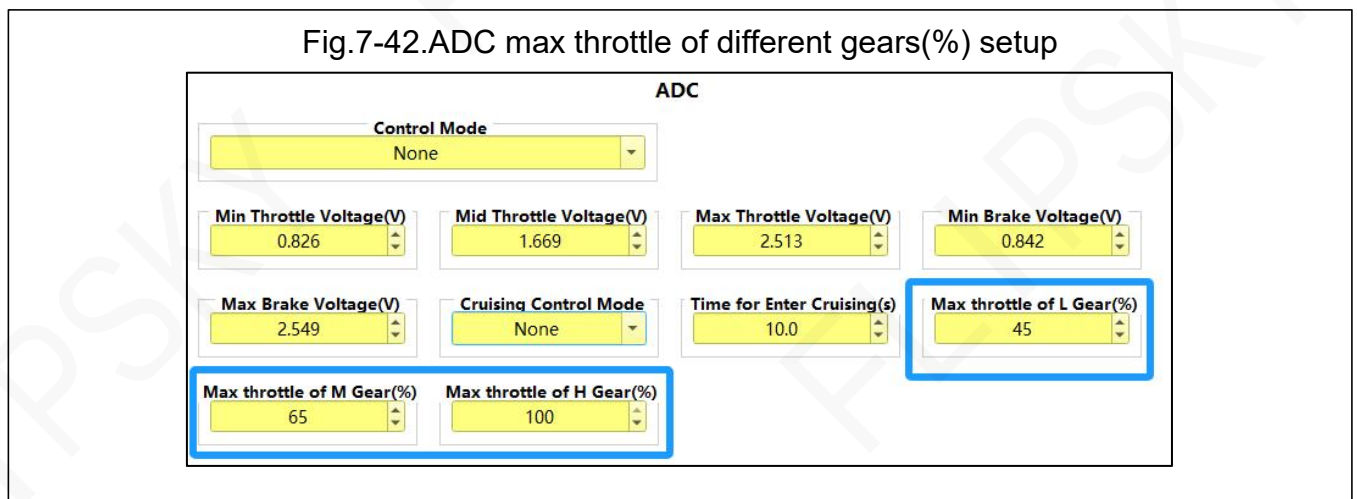
◆ **Time for Enter Cruising**

If the cruise control mode is set to Keep Throttle ADC for a While, you can set the Time for Enter Cruising per your needs. Adjustable range: 3s to 30s, and the default is 10s as shown in Fig.7-41:



7.10.4.5 ADC Max Throttle of Different Gears(%) Setup

ADC max throttle of different gears(%) setup includes “Max throttle of L Gear(%)”, “Max throttle of M Gear(%)” and “Max throttle of H Gear(%)” as Fig.7-42:



- Max throttle of L Gear(%)
- Max throttle of M Gear(%)
- Max throttle of H Gear(%)

7.10.5 PAS

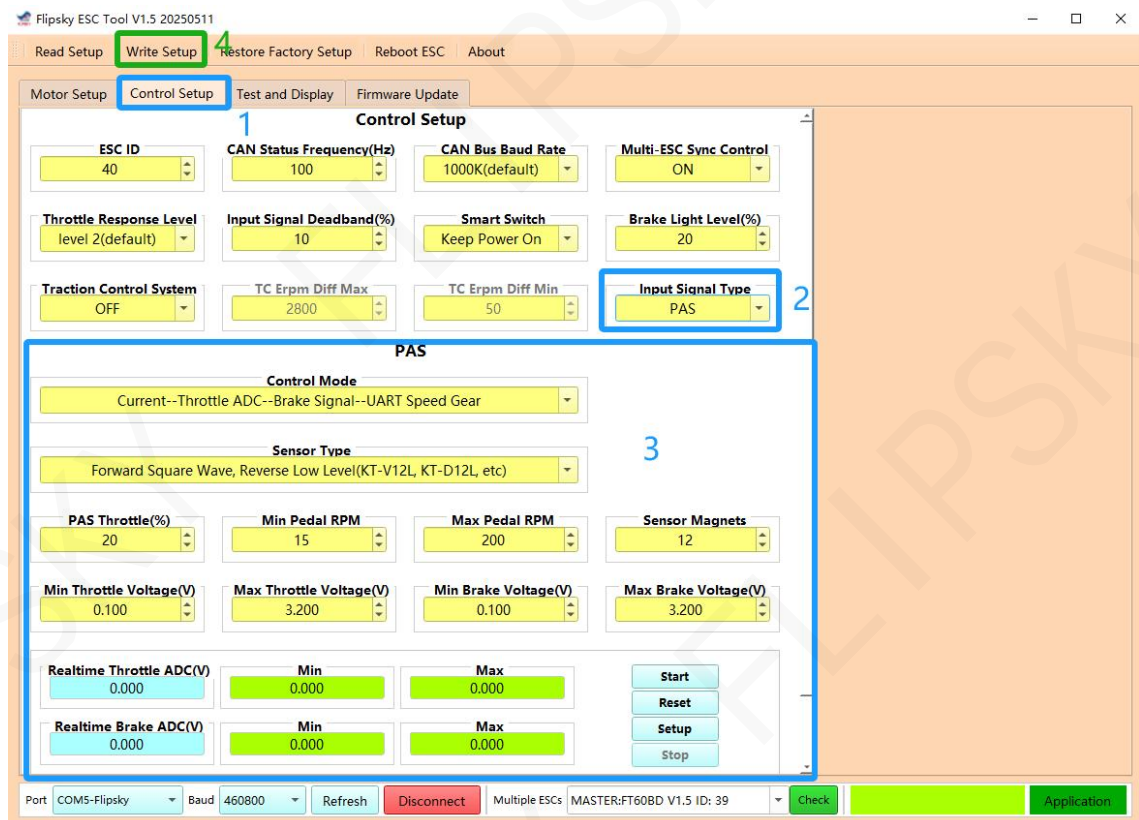
7.10.5.1 PAS Parameter Specification and Setup

PAS Function Description:

- (1) It's compatible with most of PAS cadence sensors in the market, which mainly include below three types(View in Chapter 7.8.5.3)
- (2) Enable FT ESCs to be compatible with Flipsky TFT screen display which supports real-time data like speed, voltage etc. read and PAS speed gear levels setup.
- (3) Support connect brake signal function: The ESC will cut off power output immediately once receiving the brake signal; When the control mode set to enable Brake Signal, pin GEAR_R and pin GEAR_L are used to detect the brake signal and are available at GND. That is, when the pin GEAR_R or the pin GEAR_L detects GND, the ESC will cut off the power output immediately, and disable the throttle gear.

Setup steps are as Fig.7-43 below:

Fig.7-43.PAS Input Signal Setup



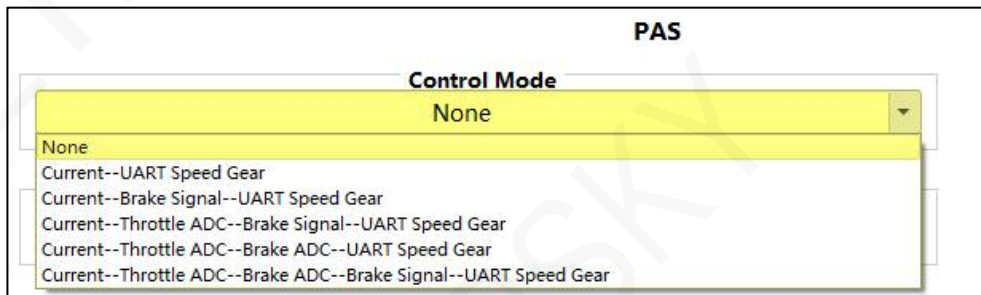
- Step 1: Click "Control Setup" to enter the setup interface
- Step 2: Switch the "Input Signal Type" to "PAS"

- Step 3: Do ADC calibration when firstly use PAS as input signal(check details in chapter 7.10.4.2)
- Step 4: Select suitable PAS Control Mode per your needs (see details in chapter 7.10.5.2)
- Step 5: Select suitable PAS Sensor Type per your needs (see details in chapter 7.10.5.3)
- Step 6: Set the PAS parameters (view details in chapter 7.10.5.4)
- Step 7: Click the upper left menu "Write Setup" to save the configuration.

7.10.5.2 PAS Control Mode Setup

Select suitable PAS Control Mode per your needs as Fig.7-44:

Fig. 7-44.PAS Control Mode



■ None: No response to any PAS input signal

■ Current--UART Speed Gear:

Current control mode: UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gears

■ Current--Brake Signal--UART Speed Gear:

Current control mode: Brake Signal--The ESC will cut off power output immediately once receiving the brake signal; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gears.

■ Current--Throttle ADC--Brake Signal--UART Speed Gear:

Current control mode: enable throttle ADC control; Brake Signal--The ESC will cut off power output immediately once receiving the brake signal; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gears.

■ Current--Throttle ADC--Brake ADC--UART Speed Gear:

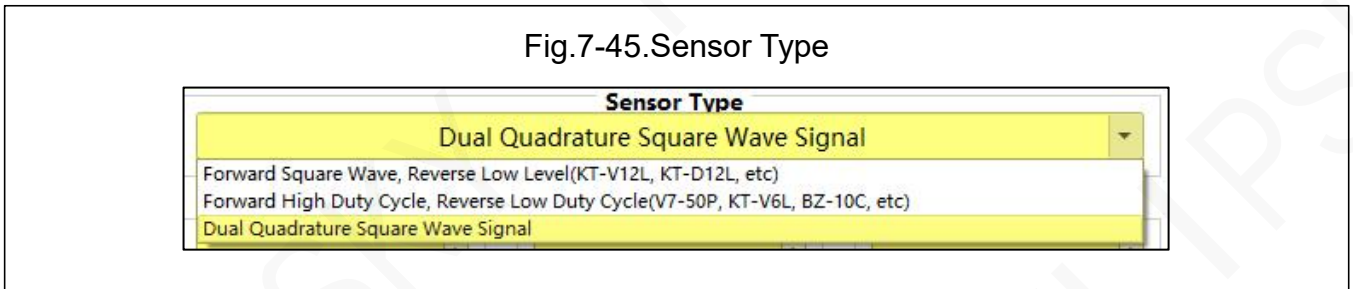
Current control mode: enable throttle ADC and Brake ADC control; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gears.

■ Current--Throttle ADC--Brake ADC--Brake Signal--UART Speed Gear:

Current control mode: enable throttle ADC and Brake ADC control; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gears.

7.10.5.3 PAS Sensor Type Setup

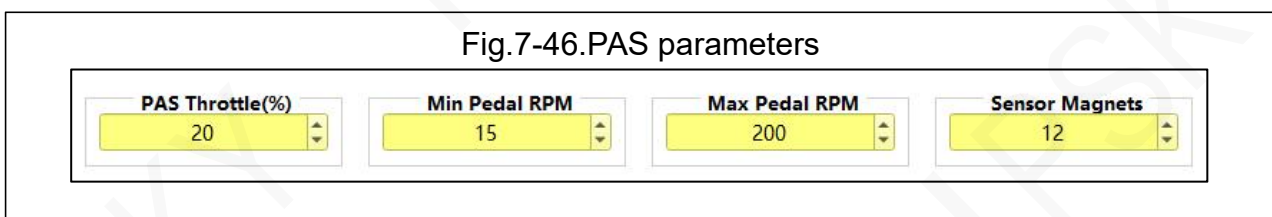
Select suitable PAS Sensor Type per your needs as Fig.7-45 below:



- **Forward Square Wave, Reverse Low Level(KT-V12L, KT-D12L, etc):**
Output square wave when rotates in forward direction; output Low Level when rotates in reverse.
- **Forward High Duty Cycle, Reverse Low Duty Cycle(V7-50P, KT-V6L, BZ-10C, etc):**
Output High duty cycle rectangular wave when rotates in forward direction; output Low duty cycle rectangular wave when rotates in reverse.
- **Dual Quadrature Square Wave Signal**

7.10.5.4 PAS Parameter Setup

PAS parameters include: “PAS Throttle”, “Min Pedal RPM”, “Max Pedal RPM” and “Sensor Magnets” as Fig.7-46:



- **PAS Throttle:** PAS max current output percentage (adjustable range: 1-95%).
The maximum PAS current output will be limited within this setting percentage of the motor current.
- **Min Pedal RPM:** the minimum PAS pedal RPM (adjustable range: 1-200RPM).
The ESC starts current output when the pedal RPM is over the mini setting value. Otherwise, the current output is zero, no current output.
- **Max Pedal RPM:** the maximum PAS pedal RPM (adjustable range: 1-500RPM).
The ESC current output will stop increasing if the pedal RPM reaches the max setting value. Once this value is exceeded, the ESC output current will remain at the maximum value (%) of

"PAS Current Throttle".

- **Sensor Magnets:** Sensor magnets q'ty adjustable range: 1-32).
PAS magnets q'ty, i.e.: 12pcs magnets have 12 pulses each turn circle.

8 Test and Display

8.1 Flipsky ESC Tool Button Control

Switch the "Input Signal Type" to "OFF" if you want to control the ESC on the Test and Display interface of Flipsky ESC Tool, only then can you control the ESC by Current, Duty, Speed, Position, Brake and other buttons as Fig.8-1:

Fig.8-1.Control button



8.1.1 Button Control Specification

Button control setup:

- Drive: Click the "Current"-current control mode (torque control mode), "Speed"- speed control mode, "Duty"-duty cycle control mode (voltage control mode) and "Position"-position control mode buttons to drive. During the driving process, the current, speed, duty cycle and position angle can be adjusted according to needs as shown in 8-2:

Fig.8-2.Four control modes

The screenshot displays the control interface of the Flipsky ESC Tool. It is divided into two main columns. The left column contains four control mode buttons, each with a numerical input field and a green button: 'Current Control(A)' with '5.0' and 'Current' button, 'Speed Control(Erpm)' with '3500.0' and 'Speed' button, 'Duty Control(%)' with '5.0' and 'Duty' button, and 'Brake Control(A)' with '5.0' and 'Brake' button. The right column contains 'Position Control(°)' with '0.00' and 'Position' button, 'Graphics Control(Hz)' with '100 Hz' and 'x1' multiplier, 'Horizontal' and 'Vertical' checkboxes (both checked), a 'Display' checkbox (unchecked), a 'Reset' button, and an 'Others' section with 'Reverse' and 'Switch' checkboxes (both checked) and a red 'Release' button. A blue box highlights the 'Current' and 'Position' buttons, and another blue box highlights the 'Speed' and 'Duty' buttons.

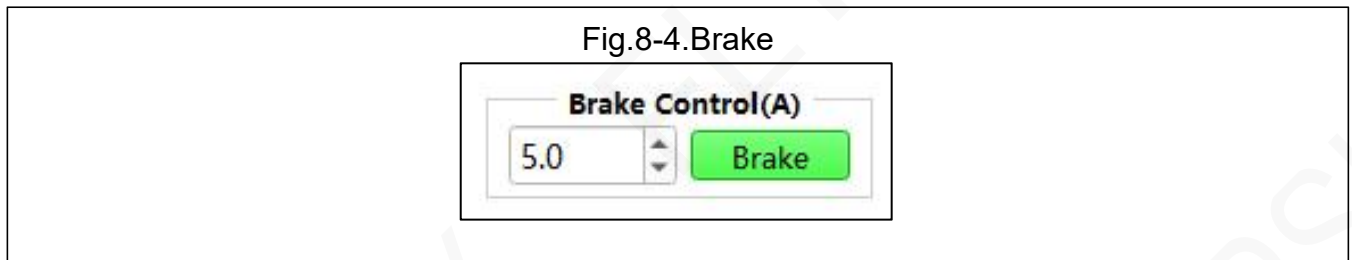
NOTE: Position Control is applicable in **Encoder Control Mode** only.

- Reverse the motor: Click "Reverse" to change the motor running direction as Fig.8-3.

Fig.8-3.Reverse direction

The screenshot shows a close-up of the 'Others' section in the control interface. It contains two checkboxes: 'Reverse' and 'Switch'. The 'Reverse' checkbox is currently unchecked and is highlighted with a red rectangular box. The 'Switch' checkbox is checked. Below these checkboxes is a red 'Release' button.

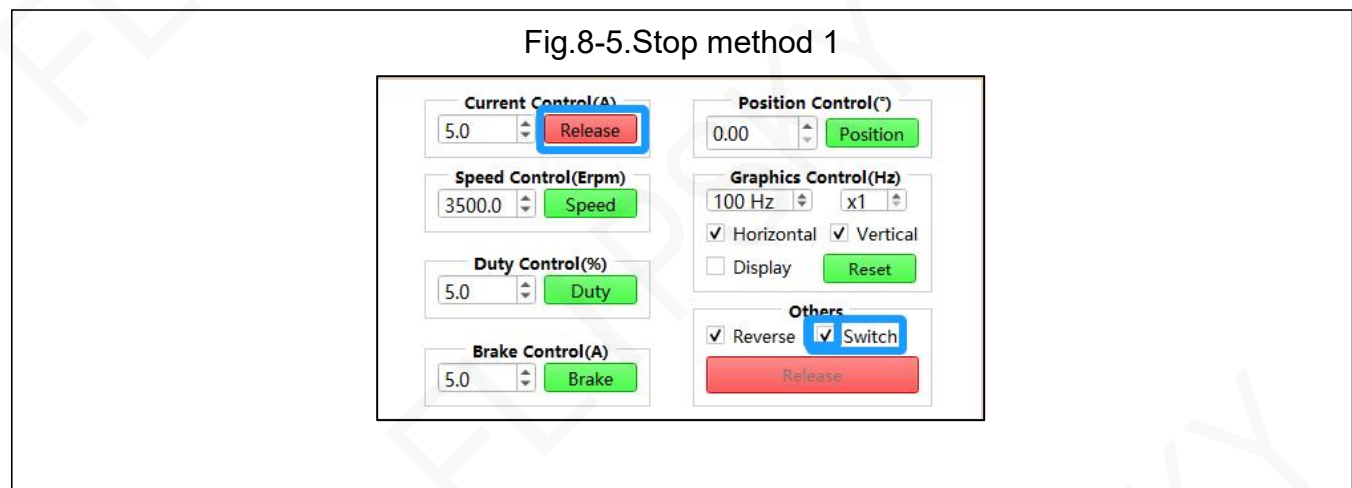
- Brake: Click "Brake" to enter the brake setup as Fig.8-4:



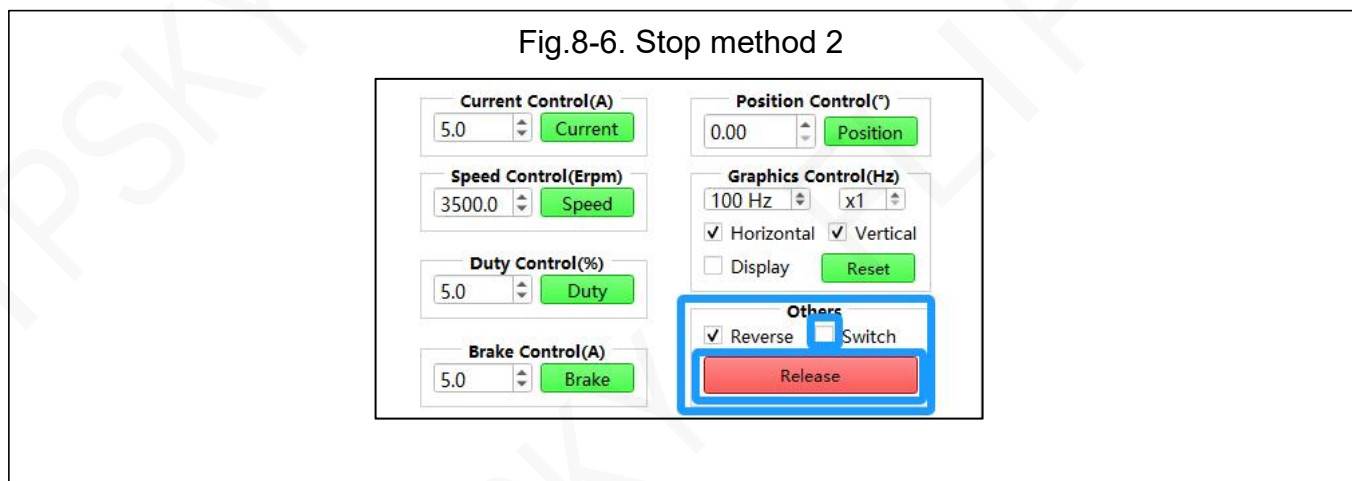
NOTE: If you use the DC power supply, please do not brake in high speed. Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed mode.

- Stop: Click "Release" to stop the motor running. Two methods available:

Click "Switch" and the "Current" under Current Control, then the "Current" will switch to "Release". Click the "Release" to stop the motor rotating as Fig.8-5:



If not tick the "Switch" and it's "Current" under Current Control, the motor can only be stopped through the "Release" under the "Others" as Fig.8-6:

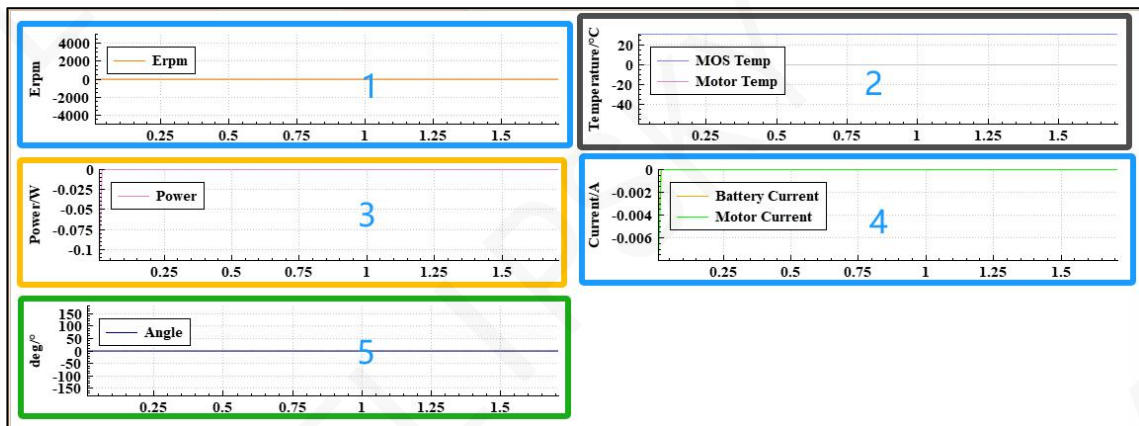


8.2 Data Analysis

8.2.1 Data Curve Display and Setup

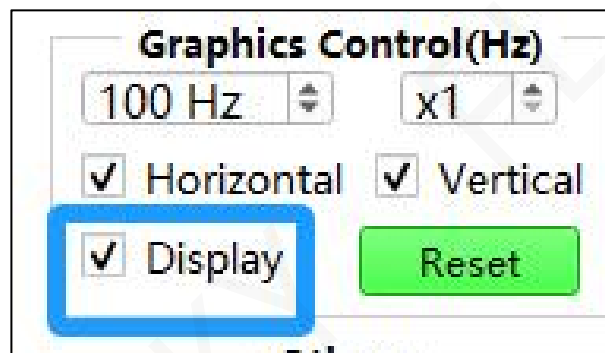
- There are 5 real-time data graphs as Fig.8-7 below:
 - Erpm curve
 - ESC MOSFET and motor temperature curve (Temperature/°C)
 - Power curve (Power/W)
 - Battery and motor current curve(Current/A)
 - 度曲线图 (deg/°) Motor deflection angle curve(deg/°)

Fig.8-7.Real-time data curve display interface



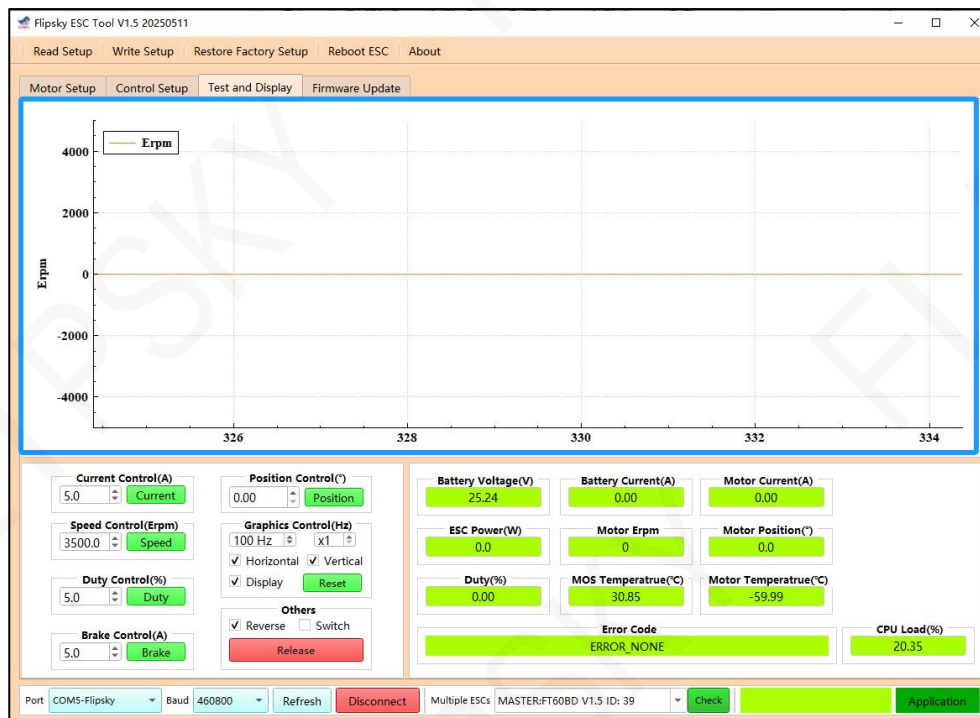
- Data Curve Display Setup Steps:
 - Activate the graph: Tick "Display" to activate graphics data display as Fig.8-8:

Fig.8-8.Graph display setup



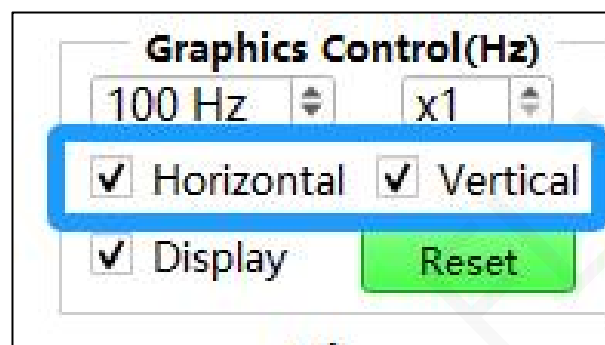
- Graph enlargement: Move the mouse cursor in the graph and left-click twice to enlarge it, and restore it in the same way as Fig.8-9 below:

Fig.8-9. Graph enlargement



- Magnify/reduce the graph horizontal and vertical axis values: Tick both "Horizontal" and "Vertical" (as Fig.8-10), move the mouse cursor into the graph and then scroll the mouse wheel to magnify/reduce.

Fig.8-10



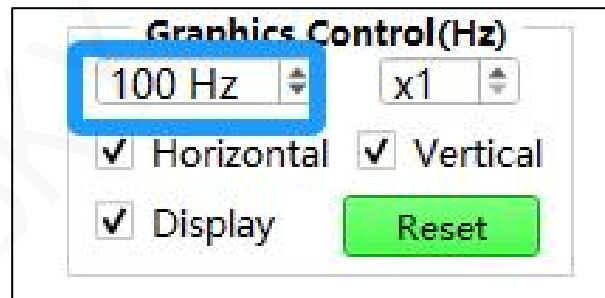
NOTE:

- (1) Tick "Horizontal" only if you just want to increase or decrease the values of horizontal axis.
- (2) Tick "Vertical" only if you just want to increase or decrease the values of vertical axis.
- (3) Tick both "Horizontal" and "Vertical" if you want to increase or decrease the values of

vertical and horizontal axis at the same time.

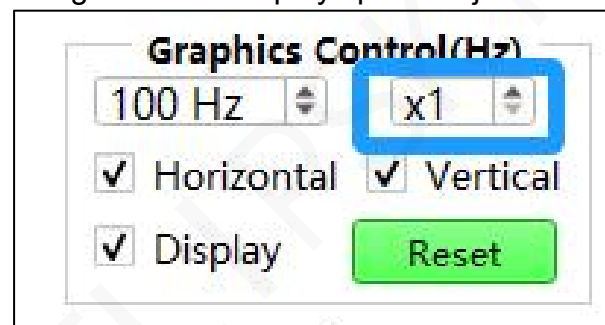
- The sampling frequency setup: default 100Hz, which means that the data samplings are 100 times per second. Adjustable setup range: 50Hz~1000HZ as Fig.8-11:

Fig.8-11. The sampling frequency setup of graphics



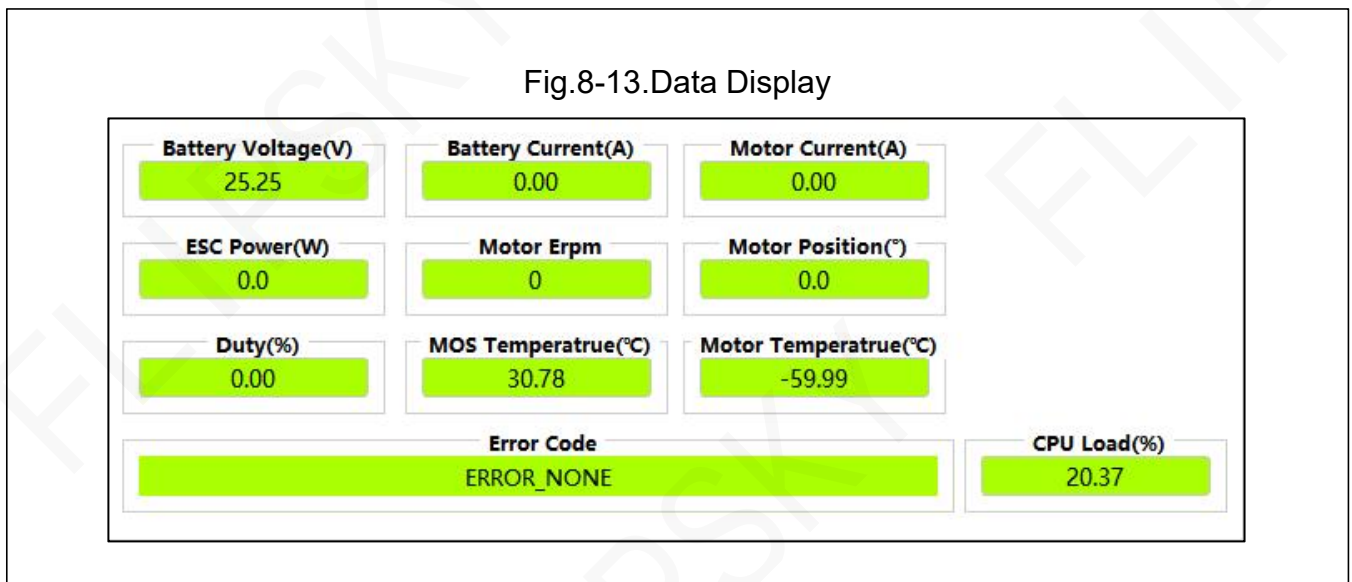
- Curve play speed setup: default "x1", adjustable range 1~10 times as8-12:

Fig.8-12. Curve play speed adjustment



8.2.2 Real-time Data Display

In order to facilitate the observation of the data and display the curve graph data in a more intuitive way, we increase some data that the curve graph does not have to better understand the equipment status as Fig.8-13:

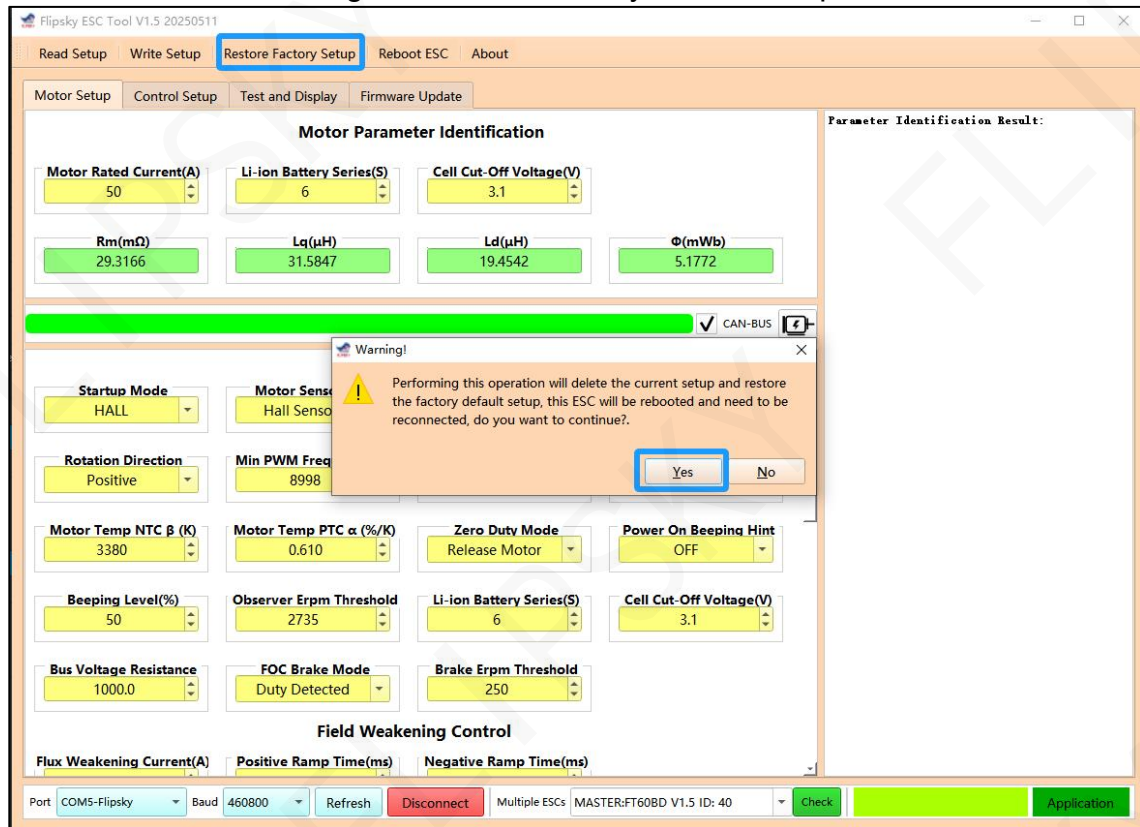


- ESC Power(W): ESC power display, corresponding to the graphic data
- Battery Voltage(V): Battery voltage display
- Battery Current(A): Battery current display, corresponding to the graphic data
- Motor Current(A): Motor current display, corresponding to the graphic data
- Motor Erpm: Motor erpm display, corresponding to the graphic data
- Motor Position: Motor deflection angle position display, corresponding to the graphic data
- Duty(%): Duty cycle percentage display
- MOS Temperature (°C): ESC MOSFET temperature display, corresponding to the graphic data
- Motor Temperature (°C): Motor working temperature display, corresponding to the graphic data
- CPU Load(%): CPU loading percentage display
- Error Code: Fault code display

9 Restore Factory Default Setup

If meet with some problems like parameters setup failure, ESCs connection failure, motor parameter identification failure etc., please try to restore the factory default setup as Fig.9-1 below:

Fig.9-1.Restore factory default setup



10 Version

Sheet 1. File Version

Date	Version	Contents Changed
2023/8/12	V1.0	
2023/11/8	V1.1	1. Fixed the problem that the Hall parameter cannot pass when the load is too heavy. 2. Optimize the parameter adjustment procedure. All interfaces are shielded during parameter adjustment to prevent other signals from affecting parameter adjustment. 3. Optimize the dead zone setting mode in ADC mode, and fix the problem that some gamepads may not start smoothly when used; 4. Rectify the fault that the electrical connection to the upper computer fails.
2024/01/19	V1.2	FT ESC Function Upgrade: 1. Add dual-drive ESC FT85BD 2. Add Smart Switch function supporting slide power on and auto turn off(support FT85BD only) 3. Add Extra Interface setup to enable front light, brake light and horn etc. for ESC FT85BD under UART control mode. Support VX4 to control the function of headlights, brake lights and horn; this function only supports FT85BD; set via Extra Interface parameters. 4. Add the function of FT series ESC under UART input signal to maintain a certain current output after releasing the throttle. In some high gear ratio (>3.5:1) skateboards, such as off-road skateboards, after releasing the throttle, there will be some resistance similar to braking, but in fact the resistance is caused by the high gear ratio. This function can maintain a certain current output after the remote releases the

		throttle, and this current can counteract these resistances, and be able to make the skateboard in an almost zero resistance release state. 5. Slow-Release Motor: Whether use this function, default as OFF 6. Slow-Decrease Time: Used for setting the continuous time of output current, default as 4.5s 7. Slow-Decrease Throttle: Used for setting the output throttle, default as 5%, which means the output current is $0.05 \times \text{Motor Current}$. 8. Slow-Decrease Duty: Only when the duty cycle is higher than this value, this function is available. 9. Add the Duty control mode of FT series ESCs in ADC mode, as well as bidirectional current and bidirectional duty cycle control modes. 10. Add Zero Duty Mode, when the user gives 0% duty cycle, the user can choose whether to keep output or release motor. 11. Repair the problem that some hub motors can not measure the magnetic chain when the inertia is relatively large. 12. Repair the problem that the maximum speed cannot limit the gear speed. Flipsk ESC Tool Function Upgrade: 1. Repair the abnormal problem that the curve cannot display the waveform corresponding to the waveform name. 2. Optimize the version judgment to prevent user from using unmatched firmware and causing abnormal problems. 3. Repair the problem that sometimes real-time data cannot be displayed normally after the Flipsky ESC Tool turns on.
2024/04/22	V1.3	FT ESC Function Upgrade: 1. Add FT ESC FT85RS 2. Optimize the motor flux linkage detection for some small motors with large resistance. 3. Optimize real-time data acquisition protocol (Notes: the remote VX4 needs to update firmware V1.3.0, otherwise the real-time data will display abnormally and the parameter adjustment results will display abnormally.)

		4. The minimum set voltage of lithium battery can be set to 2.6V to support lithium iron phosphate battery. 5. Add PAS Function: (1) It's compatible with most of PAS cadence sensors in the market, which mainly include below three types: - Forward Square Wave, Reverse Low Level like KT-V12L, KT-D12L etc. - Forward High Duty Cycle, Reverse Low Duty Cycle like V7-50P, KT-V6L, BZ-10C etc. - Dual quadrature square wave signal output: the A-phase signal leads the B-phase signal when rotating in the forward direction, while the A-phase signal lags behind the B-phase signal when rotating in the backward / reverse direction. (2) Enable FT ESCs to be compatible with Flipsky TFT screen display which supports real-time data like speed, voltage etc. read and PAS speed gear levels setup. (3) Support Brake Signal function: The ESC will cut off power output immediately once receiving the brake signal; When the control mode set to enable Brake Signal, pin GEAR_R and pin GEAR_L are used to detect the brake signal and are active at Vol. That is, when the pin GEAR_R or the pin GEAR_L detects a Vol(GND), the ESC will cut off the power output immediately, and disable the throttle gear. Flipsky ESC Tool Function Upgrade: - Add FT85RS ESC - Add PAS function
2024/8/24	V1.4	FT ESC Function Upgrade: - Add FT60BD ESC - Optimize the motor voltage detection accuracy by adding motor "Bus Voltage Resistance" setup. - Add Flux Weakening Current Control. - Add encoder function. Enable encoder control setup like AS5047 (SPI)

		and ABI Encoder. 5. Add speed control PID mode. Maintain the default parameters in most situations. 6. Add position control PID mode (for encode control status only). Maintain the default parameters in most situations. 7. Add more types of motor temperature sensors such as NTC-10K-25℃, NTC-100K-25℃, PT1000, PTC-1K-100℃, KTY83 series, KTY84 series etc. 8. Add ESC power on beeping alert ON/OFF setup. Also, the beeping voice level percentage can be adjusted. 9. Add Brake Light Level setup to enable the light brightness to be adjusted during the braking process (available for FT60BD and FT85BD only at present). 10. In UART control mode, add Assisted Turnaround Mode setup like fixed-circle turnaround for 2WD mode and tank turnaround for 4WD mode. 11. In UART control mode, add Brake-Parking Mode ON/OFF setup to support parking mode by pulling remote controller brake gear/bar (achieved by transferring current into the motor D axis) 12. In UART control mode, add Off-Road Climbing Mode ON/OFF setup to switch duty cycle control and allow users to pass off-road climbing process easily by controlling speed of the four wheels synchronously 13. In ADC control mode, add Cruising Control Mode setup in three different situations. (1) Self-locking switch (2) Reset switch (3) Throttle hold Flipsky ESC Tool Function Upgrade: 1. Add FT60BD ESC. 2. Add erpm control and position control modes in Test and Display interface; add position data display.
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2025/5/11	V1.5	FT ESC Function Upgrade 1. Add firmware of FT85RS with enclosure. Enclosure version and water cooling version use the same firmware 2. Add firmware of FT85GS with enclosure 3. Add firmware of DR60BS UAV ESC 4. Optimize the brake control algorithm and added three new brake modes control: (1) When entering the brake state and the duty cycle is detected to be lower than 0.001, it switches to three-phase short circuit state; (2) When entering the braking state and the detected erpm is lower than the Brake Erpm Threshold value, it will switch to the three-phase short-circuit state; (3) When entering the braking state, it will switch to the three-phase short-circuit state when it detects the duty cycle is lower than 0.001 and the absolute value of erpm is lower than the Brake Erpm Threshold value at the same time; 5. Optimize Flux Weakening Control Algorithm, add new Flux Weakening Control Current Positive Ramp Time and Negative Ramp Time. Users can adjust the Flux Weakening Current Response Time by the value of Positive Ramp Time and Negative Ramp Time. For example, when Positive Ramp Time is set to 1000ms and Flux Weakening Current is 20A, then when the duty cycle reaches 90% or more, the Flux Weakening Current will increase from 0A to 20A within 1000ms, and when Negative Ramp Time is set to 1000ms, when the current Control command is set to 0, then the weak magnetic current will decrease from 20A to 0A within 1000ms, and then end the weak magnetic control. 6. Add multiple ESCs jointly control to realize TRC control. This algorithm can be used to prevent the driving wheels from slipping when multiple ESCs accelerate the driving motors at the same time, and this mode is only for the current control (does not support the current mixing bi-directional control in the PPM mode); 7. Add CAN Status Frequency parameter. Each ESC will automatically return real-time data through the CAN bus, which includes motor current, battery current, motor electronic speed, duty
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		cycle, MOSFET temperature, motor temperature, battery voltage data. The real-time data above will be sent back to the CAN bus at regular intervals. The sending frequency can be set via CAN Status Frequency. 8. Add unidirectional speed PID control, bidirectional speed PID control and bidirectional current mixing control under PPM mode. The maximum speed value of the speed PID control mode in PPM mode can be set by the parameter PID Maximum Erpm. For bidirectional current hybrid control, the control logic is as follows: (1) When the rocker has no movement and is centered (PPM input signal is Mid Pulse Width value), there is no current motor power output; (2) When you push the throttle forward (PPM input signal > Mid Pulse Width value), then the motor turns positively; (3) When you push the throttle forward, the motor is rotating forward, if you pull back the rocker now (the input signal < Mid Pulse Width value), then the motor enters the braking state; (4) When the motor enters the braking state, after the rocker is centered, pull back the throttle again, the motor will enter the reverse drive state. Only when you push the throttle forward again to make the motor rotate positively, and then pulls the throttle backward, then the motor will re-enter the braking state. Bi-directional Current Mixing Control Mode is suitable for RC Car. 9. Add unidirectional speed PID control, bidirectional speed PID control and bidirectional speed PID control with reverse gear in ADC mode. 10. Add maximum throttle percentage for different gears in ADC mode, the maximum throttle percentage for L, M and H gears can be set by the parameters Max throttle of L Gear, Max throttle of M Gear and Max throttle of H Gear respectively.